τ²-bench failure analysis: English vs Bangla (gemma4:31b and gpt-oss:20b)

Generated by `analysis/tau2/fa_report.py` on 2026-10-10, following `failure_analysis_pipeline.md` step by step. Every number below is recomputed from the paired run table (`analysis/tau2/paired_current.pkl`, rebuilt from every `results.json`), the automatic per-run signals (`fa_signals.py`) and the 66 EN-pass/BN-fail trace notes in `tau2_refresh_trace_notes_all.csv`.

Summary of results

- **gemma4:31b is worse in Bangla.** 49.5% in English against 38.1% in Bangla on 218 paired tasks (gap +11.5 pp, 95% interval [5.0, 17.4] pp, McNemar p = 0.001); the teammate set of 219 different tasks agrees (+8.7 pp, p = 0.034).
- **gpt-oss:20b shows no Bangla cost** (38.9% vs 39.6%, p = 1.000). On the 149 tasks shared by all four cells the models differ by +14.1 pp (interval [2.0, 26.2] pp).
- **Most Bangla-only failures come from the Bangla user simulator, not the agent.** Of the 66 EN-pass/BN-fail cases with trace notes, 35 (53%) are user-simulator errors: the simulated user confirms and ends the chat in one message, gives an internal ID as a name, answers in English, or narrates a phone action instead of performing it.
- **Genuine Bangla agent failures are few:** 8 cases (6 gemma, 2 gpt-oss), mostly entity grounding and telecom tool use; 21 more are general weaknesses that also occur in English.
- **The gemma gap shrinks when the simulator artifact is removed symmetrically:** +11.5 pp as scored, +7.1 pp after dropping premature-stop dialogues in either language (p = 0.090), +5.8 pp with the Bengali-digit grader fix as well (p = 0.175).
- **Limits.** One trial per cell; the premature-stop detector is heuristic; 20 of the notes are single-annotator; the teammate set has no traces; gpt-oss retail and banking are mostly excluded by provider errors.

0. Scope and how the pipeline was adapted

This is the benchmark the pipeline was written for: **τ²-bench, customer-service dialogues in which an LLM user simulator plays the customer and the agent must change a database through tools**. Domains airline, retail, telecom and banking (mock is excluded everywhere). One English and one Bangla run per task, one trial, Ollama, agent and user simulator are the same model. Pass = reward ≥ 1 (database state, required actions, required information and, in telecom, the user's device state). Runs that ended in an infrastructure or quota error are dropped on both sides.

Pipeline item	In this report
Dataset	One: τ²-bench. Domain = airline, retail, telecom, banking.
Model	gemma4:31b: primary set 218 paired tasks with full traces, plus the teammate set (219 paired tasks, results only, no traces). gpt-oss:20b: 149 paired tasks. 149 tasks are gradeable in all four cells.
"Bangla user simulator"	Present and important: in the Bangla cells the user simulator speaks Bangla. Confound C2 is counted separately.
Case selection (§3)	Every EN-pass/BN-fail case of the primary sets has a trace note (40 gemma, 26 gpt-oss). The 46 teammate EN-pass/BN-fail cases have no local traces and are counted only.
Every failed run	Not hand-coded. Every failed run gets an **automatic signal** from its grading record and messages (how it ended, which check failed, whether the simulator stopped before the agent acted); Figure 8.
EN-fail/BN-pass side	Not hand-coded. Symmetric treatment uses the automatic premature-stop detector on both languages (Section 11, S6).
Second reviewer	46 of the 66 notes come from the earlier adjudicated set (two reviewers); the 20 newest notes are single-annotator (Section 14).

R. Results of all runs

Model	Prompt	Set	Tasks	All	Airline	Retail	Telecom	Banking
gemma4:31b	English	primary	218	108 / 218 (49.5%)	33 / 50 (66.0%)	30 / 41 (73.2%)	41 / 87 (47.1%)	4 / 40 (10.0%)
gemma4:31b	Bangla	primary	218	83 / 218 (38.1%)	23 / 50 (46.0%)	21 / 41 (51.2%)	36 / 87 (41.4%)	3 / 40 (7.5%)
gpt-oss:20b	English	primary	149	58 / 149 (38.9%)	19 / 48 (39.6%)	18 / 41 (43.9%)	18 / 46 (39.1%)	3 / 14 (21.4%)
gpt-oss:20b	Bangla	primary	149	59 / 149 (39.6%)	22 / 48 (45.8%)	11 / 41 (26.8%)	23 / 46 (50.0%)	3 / 14 (21.4%)
gemma4:31b	English	teammate 250	219	74 / 219 (33.8%)	–	20 / 33 (60.6%)	53 / 130 (40.8%)	1 / 56 (1.8%)
gemma4:31b	Bangla	teammate 250	219	55 / 219 (25.1%)	–	18 / 33 (54.5%)	37 / 130 (28.5%)	0 / 56 (0.0%)

Infrastructure-error pairs are excluded before counting; this is why the denominators differ by domain and model (gpt-oss retail and banking lost most tasks to provider quota and to the `gpt-4.1` judge that is not available on Ollama).

R.1 The study in two figures



Figure 1. The pipeline, step by step: what each step does, how, and what it hands to the next.

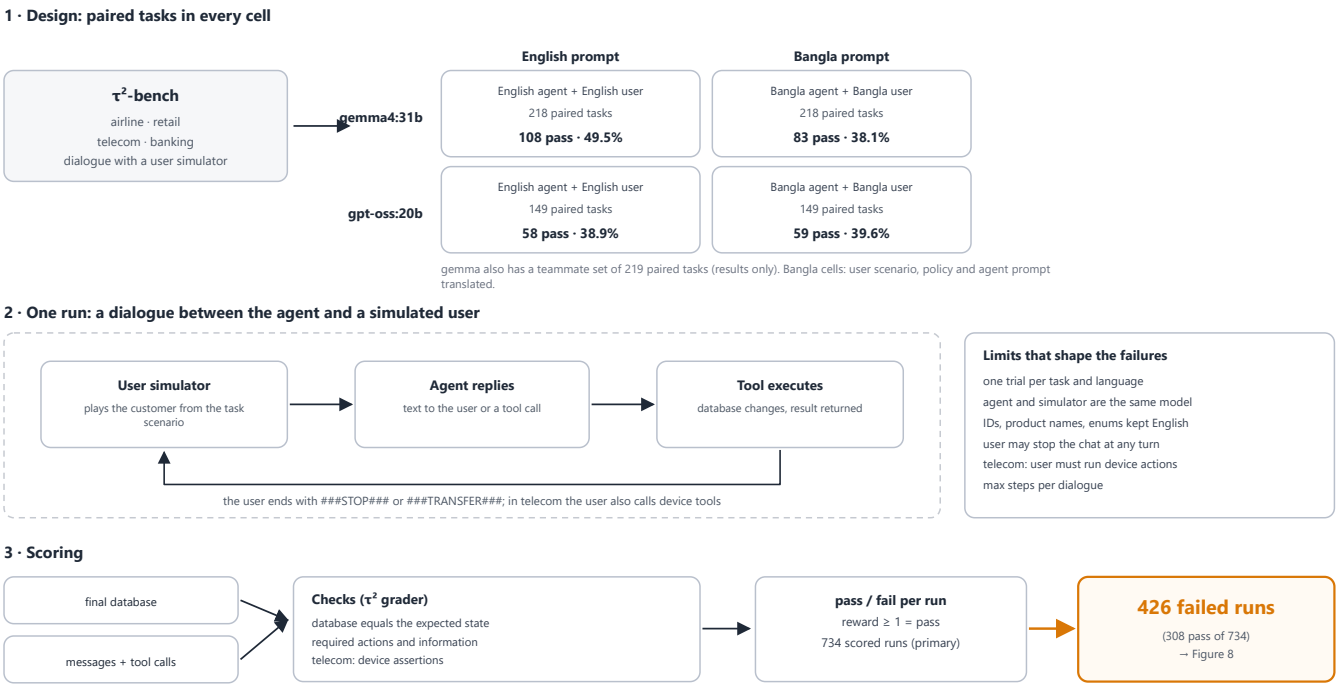


Figure 2. How the study works: the four cells, one dialogue, and the scoring.

R.2 Success rate by run and domain

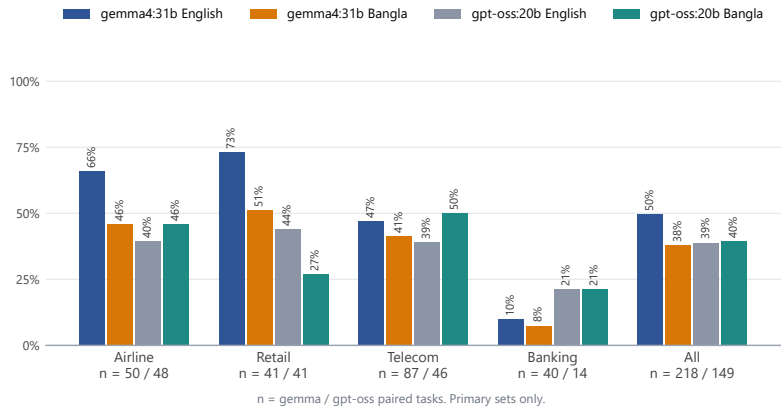
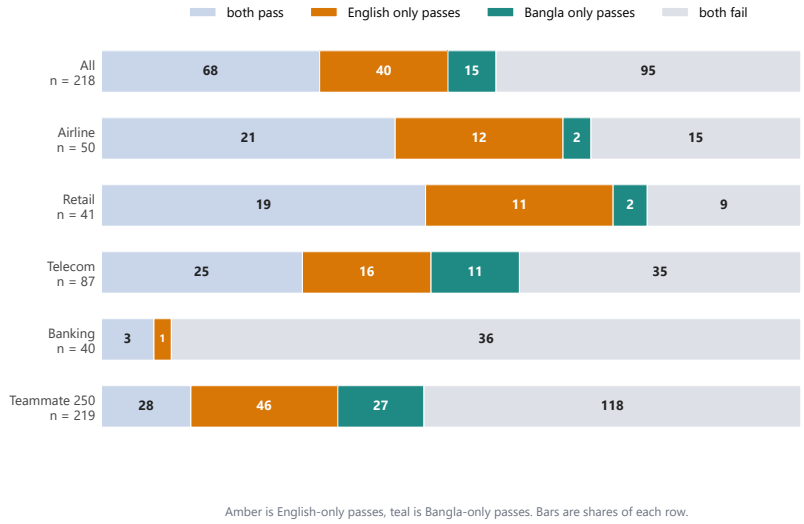


Figure 3. Success rate of the four runs by domain.

- gemma4:31b passes **49.5%** in English and **38.1%** in Bangla (gap +11.5 pp, 95% interval [5.0, 17.4] pp, McNemar p = 0.001).
- gpt-oss:20b passes **38.9%** and **39.6%** (gap -0.7 pp, interval [-10.7, 8.7] pp, p = 1.000): no Bangla cost.
- On the 149 tasks gradeable in all four cells the gemma gap is +13.4 pp and the gpt-oss gap -0.7 pp; the models differ by **+14.1 pp** (95% interval [2.0, 26.2] pp), so the Bangla cost is model-dependent.
- gemma's teammate set (219 tasks, different tasks) shows the same direction: 33.8% vs 25.1% (p = 0.034); pooled with the primary set, 41.6% vs 31.6% on 437 tasks (MELR 24.2%, p = <0.001).

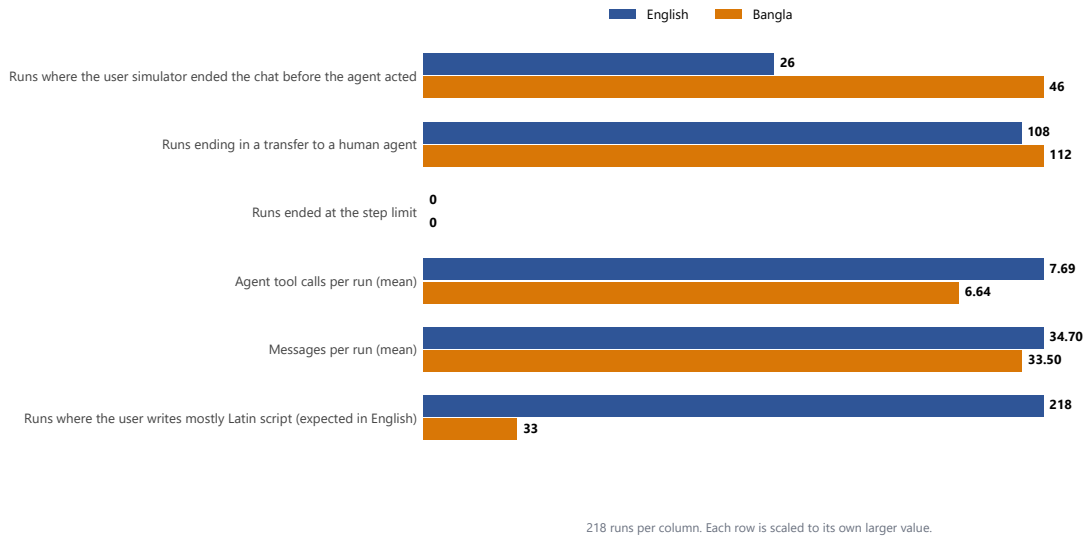
- Banking is near zero in both languages for both models, so it carries no language signal.

R.3 gemma4:31b, English against Bangla, task by task



Amber is English-only passes, teal is Bangla-only passes. Bars are shares of each row.

Figure 4. Paired outcome of gemma4:31b English and Bangla on the same tasks.

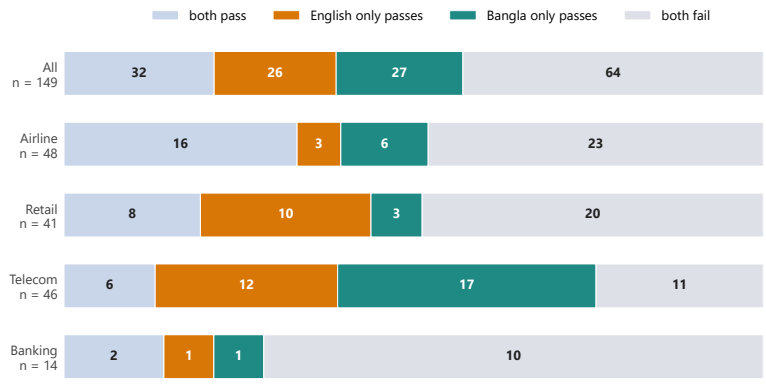


218 runs per column. Each row is scaled to its own larger value.

Figure 5. gemma4:31b English against Bangla: how dialogues end and how the user simulator behaves.

- In **46** gemma Bangla runs the simulated user confirmed and ended the chat (###STOP### or ###TRANSFER###) in the same message, before the agent could make a required write; in English this happens in 26 runs. The detector is the one in `regrade_stop.py`; a hand check found about half of its flags are true premature stops (Section 11).
- In 33 Bangla runs the simulated user wrote mostly in Latin script (English), a simulator fidelity problem the agent cannot fix.

R.4 gpt-oss:20b, English against Bangla, task by task



Amber is English-only passes, teal is Bangla-only passes. Bars are shares of each row.

Figure 6. Paired outcome of gpt-oss:20b English and Bangla on the same tasks.

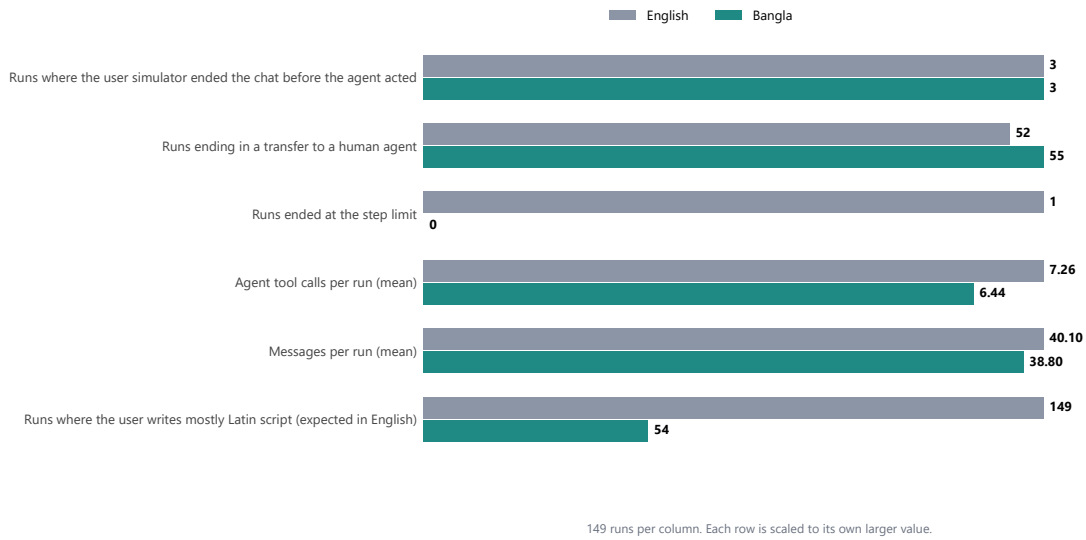


Figure 7. gpt-oss:20b English against Bangla: how dialogues end and how the user simulator behaves.

R.5 Why runs fail

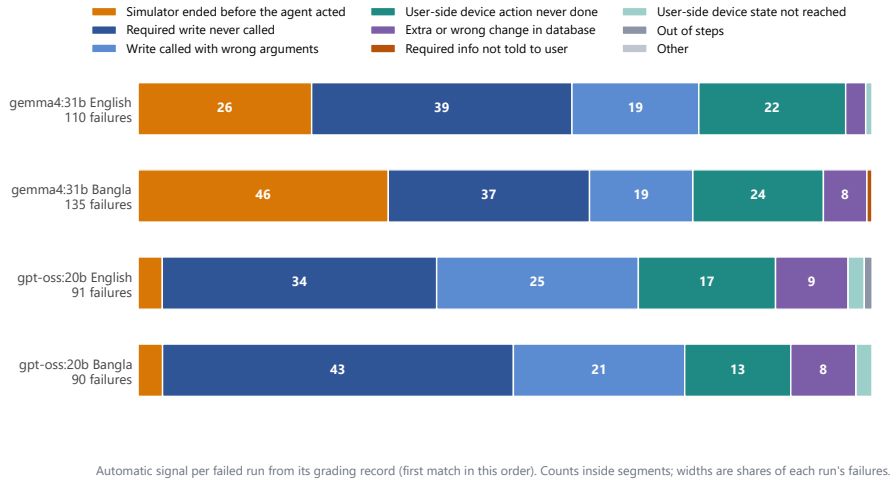


Figure 8. Automatic failure signal of every failed run, for the four primary runs.

- **The simulator ending the chat early is the largest Bangla-side increase for gemma**: 46 of 135 Bangla failures against 26 of 110 English failures.
- **A required write that is never called** and **a write with wrong arguments** are the other large families, in both languages: the agent's own errors are largely shared.
- **Telecom adds user-side failures**: the simulated user must toggle settings on a simulated phone; when it narrates instead of calling the device tool, the device check fails.

R.6 What fails everywhere



Figure 9. How the four runs agree on each of the 149 tasks gradeable in all four cells. Filled dot = passes, open dot = fails.

- **33** tasks fail in all four runs and **21** pass in all four.
- **11** tasks pass in English and fail in Bangla for **both** models: the most direct candidates for a Bangla-side cause shared across models.

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1. Data prepared (pipeline section 1)

Every task has one row per model. Alignment check: rows are keyed by (model, domain, task ID); for every key one English and one Bangla result exist (218 gemma, 149 gpt-oss after dropping infrastructure errors). Rule when a task ran more than once: prefer a non-error result, then the latest run (`build_current.py`).

Field	Where it comes from
dataset, model, task ID, run ID	τ^2-bench / agent LLM in `results.json` / `task_id` / run folder (`en100_*` , `bn_inc_*` , `oss_*` , `*_fix_*`)
English and Bangla pass/fail	`reward_info.reward` ≥ 1; error terminations excluded → `analysis/tau2/paired_current.pkl`
Trajectory paths	`tau2-bench/data/simulations/<run>/results.json` (EN), `tau2-bench-bn/data/simulations/<run>/results.json` (BN)
Final state, checks	`reward_info` : `db_check` , `action_checks` , `env_assertions` , `communicate_checks`
Automatic signals	`analysis/tau2/fa_signals.py` → `fa_signals.pkl`
Trace notes	`analysis/tau2/tau2_refresh_trace_notes_all.csv` (66 rows; rendered traces of the newest 20 in `new_case_traces/`)
Teammate 250	results only (`full_current.pkl` , source = teammate); no traces in this repository

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2. Basic outcome (pipeline section 2)

Model	Setting	n	EN pass	BN pass	Gap (EN−BN)	MELR	PP	PF	FP	FF
gemma4:31b	Overall	218	108 / 218 (49.5%)	83 / 218 (38.1%)	+11.5 pp	23.1%	68	**40**	15	95
	Airline	50	33 / 50 (66.0%)	23 / 50 (46.0%)	+20.0 pp	30.3%	21	**12**	2	15
	Retail	41	30 / 41 (73.2%)	21 / 41 (51.2%)	+22.0 pp	30.0%	19	**11**	2	9
	Telecom	87	41 / 87 (47.1%)	36 / 87 (41.4%)	+5.7 pp	12.2%	25	**16**	11	35
	Banking	40	4 / 40 (10.0%)	3 / 40 (7.5%)	+2.5 pp	25.0%	3	**1**	0	36
gpt-oss:20b	Overall	149	58 / 149 (38.9%)	59 / 149 (39.6%)	-0.7 pp	-1.7%	32	**26**	27	64
	Airline	48	19 / 48 (39.6%)	22 / 48 (45.8%)	-6.2 pp	-15.8%	16	**3**	6	23
	Retail	41	18 / 41 (43.9%)	11 / 41 (26.8%)	+17.1 pp	38.9%	8	**10**	3	20
	Telecom	46	18 / 46 (39.1%)	23 / 46 (50.0%)	-10.9 pp	-27.8%	6	**12**	17	11
	Banking	14	3 / 14 (21.4%)	3 / 14 (21.4%)	+0.0 pp	0.0%	2	**1**	1	10
gemma4:31b	Teammate 250	219	74 / 219 (33.8%)	55 / 219 (25.1%)	+8.7 pp	25.7%	28	**46**	27	118
gemma4:31b	Primary + teammate	437	182 / 437 (41.6%)	138 / 437 (31.6%)	+10.1 pp	24.2%	96	**86**	42	213

MELR = (EN − BN) / EN. PP = EN-pass/BN-pass, **PF = EN-pass/BN-fail**, FP = EN-fail/BN-pass, FF = EN-fail/BN-fail.

3. Which cases were analysed (pipeline section 3)

Model	Group	Count	How analysed here
gemma4:31b	EN-pass / BN-fail	40	**All 40** read trace-against-trace and labelled (Section 4, Appendix A).
	EN-fail / BN-pass	15	Automatic signals on the English failure; symmetric premature-stop filter (Section 11).
	EN-fail / BN-fail	95	Automatic signals on both sides (Figure 8, Appendix B): the general-difficulty baseline.
	EN-pass / BN-pass	68	Counted only.
gpt-oss:20b	EN-pass / BN-fail	26	**All 26** read trace-against-trace and labelled (Section 4, Appendix A).
	EN-fail / BN-pass	27	Automatic signals on the English failure; symmetric premature-stop filter (Section 11).
	EN-fail / BN-fail	64	Automatic signals on both sides (Figure 8, Appendix B): the general-difficulty baseline.
	EN-pass / BN-pass	32	Counted only.
gemma4:31b (teammate)	EN-pass / BN-fail	46	Counted only: no traces available locally.

4. Trace notes (pipeline section 4)

The failure analysis follows one procedure for every EN-pass/BN-fail case (Figure 10 shows it on a real task; Figure 11 is the coding scheme). One note per case, with the fields the pipeline asks for. All notes are in Appendix A. Two worked examples show the level of detail.

airline 19 · gemma4:31b · C2

- **Task goal:** You will have a crazy half-day trip to Texas. It is in your reservations but you don't remember the reservation id. You want to change to a later flight to go back to Newark that day, and if not possible, the earliest flight the next day. Your current return flight departs 3pm (on the 28th).
- **English:** Agent refused to modify basic economy, user asked to cancel with insurance, agent called cancel_reservation after confirmation; passed.
- **Bangla:** Agent asked to confirm cancelling Z7GOZK; the Bangla user confirmed and appended ###STOP### in the same message, ending the dialogue before cancel_reservation was called.
- **First divergence:** Final BN user turn combined confirmation with ###STOP###, so the agent never got to execute the cancellation.
- **Final failed / missing action:** cancel_reservation(Z7GOZK) never executed
- **First failure point:** Premature termination · **Type:** Dialogue-state failure
- **Confound:** Bangla user-simulator error (high)
- **Evidence:** [USER] হ্যাঁ, আমি নিশ্চিত। দয়া করে আমার রিজার্ভেশন Z7GOZK বাতিল করে দিন। ###STOP###

airline 9 · gemma4:31b · C1

- **Task goal:** You want to cancel two of your upcoming reservations (IFOYYZ and NQNU5R) and change a third (M20IZO) to a nonstop flight if available.
- **English:** Agent noticed NQNU5R flights (May 13-14) already flown, refused cancel, searched JFK-MCO, and transferred; reward 1.
- **Bangla:** Agent claimed NQNU5R can be cancelled since business class, got the user to confirm, and called cancel_reservation on an already-flown reservation.
- **First divergence:** BN agent failed to check the flight dates against the policy (flown flights cannot be cancelled), unlike EN which flagged it.
- **Final failed / missing action:** Must NOT cancel NQNU5R (flights already departed); BN called cancel_reservation(NQNU5R)
- **First failure point:** Planning · **Type:** Wrong tool/action
- **Confound:** Genuine Bangla-related agent failure (high)
- **Evidence:** cancel_reservation({"reservation_id": "NQNU5R"}) after "এটি একটি বিজনেস ক্লাস ফ্লাইট, তাই এটি বাতিল করা সম্ভব"

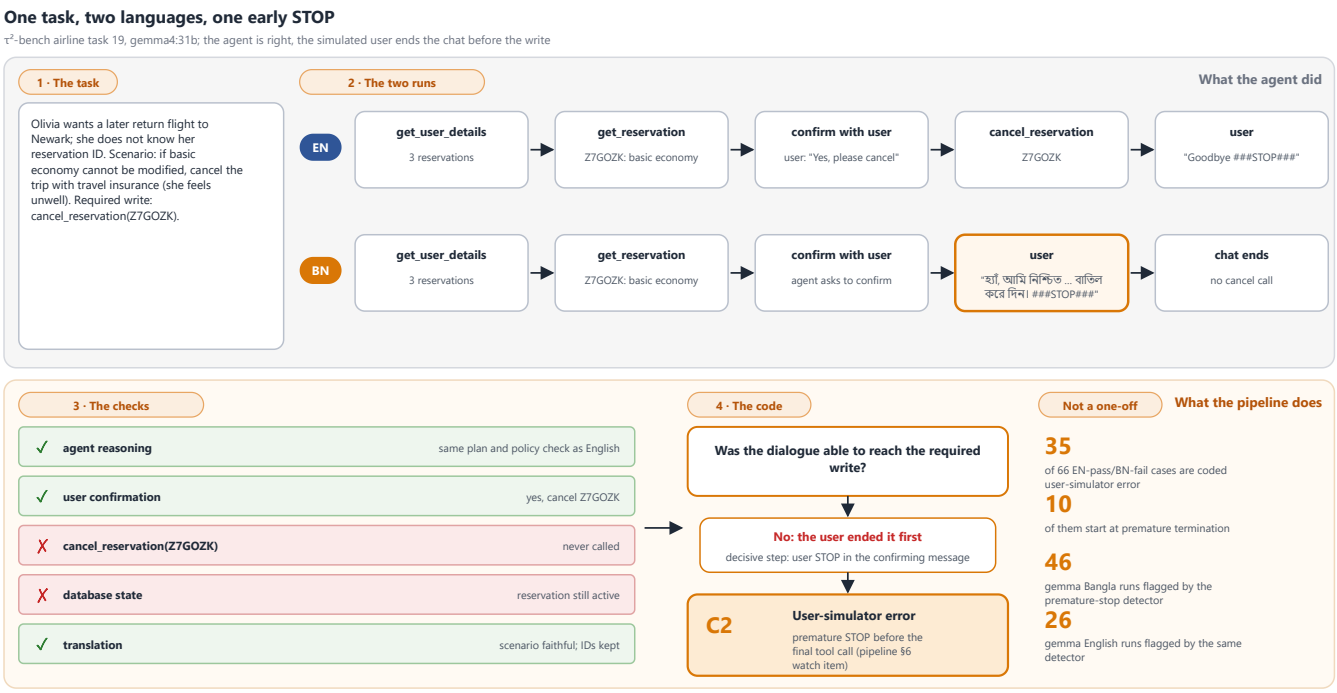


Figure 10. One real task carried through the pipeline: the agent did everything right, but the Bangla user simulator confirmed and stopped in the same message.

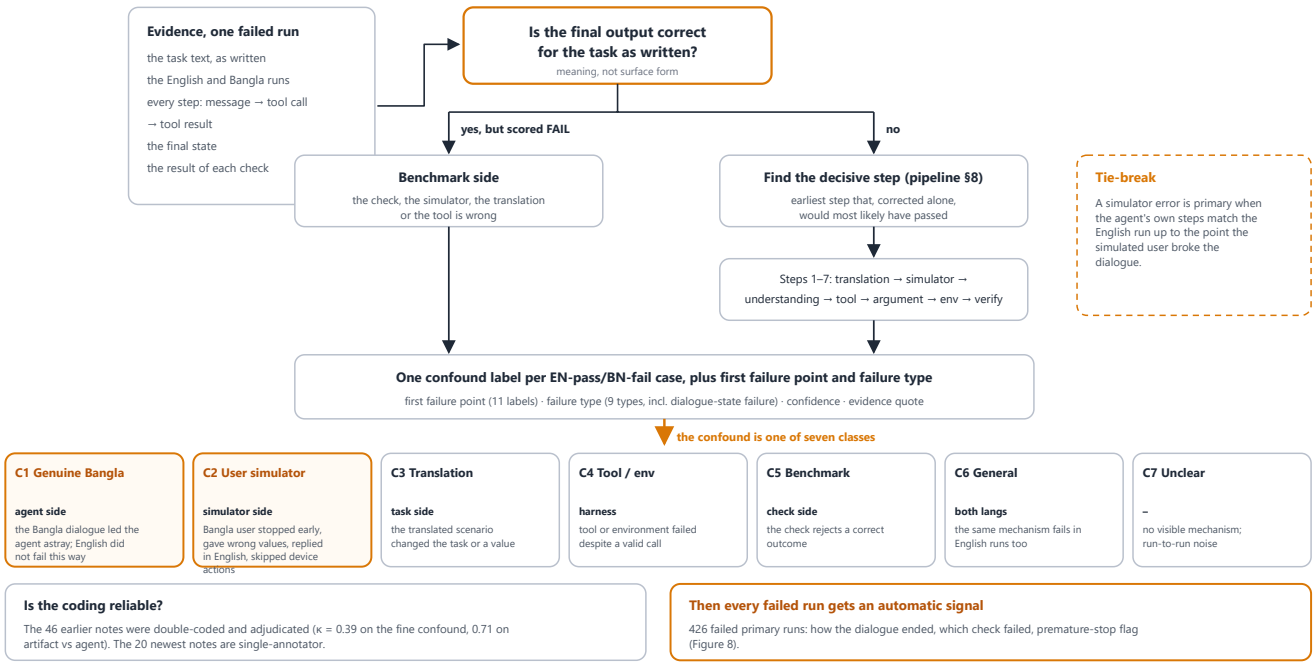


Figure 11. How failures are characterized: one question, one decisive step, one confound class per case.

5. First failure point (pipeline section 5)

Rule applied: label the earliest upstream point, not the visible final error. When the simulated user broke the dialogue, the first failure point is where the dialogue broke (often premature termination), not the agent's last turn.

First failure point	gemma	share of 40	gpt-oss	share of 26	both
Instruction understanding	0	0.0%	3	11.5%	3
Entity or slot grounding	7	17.5%	5	19.2%	12
Task constraint preservation	4	10.0%	3	11.5%	7
Planning	6	15.0%	5	19.2%	11
Tool selection	6	15.0%	1	3.8%	7
Tool-argument construction	4	10.0%	2	7.7%	6
Result verification	1	2.5%	3	11.5%	4
Premature termination	11	27.5%	1	3.8%	12
Final response only	1	2.5%	0	0.0%	1
Unclear	0	0.0%	3	11.5%	3

Reading the table: premature termination, entity grounding and planning lead. Tool syntax is never the first failure point: the Bangla failures are about what the dialogue carries (names, constraints, when it ends), not about how tools are called.

6. Failure type (pipeline section 6)

Failure type	gemma	share of 40	gpt-oss	share of 26	both
Understanding failure	5	12.5%	4	15.4%	9
Planning failure	6	15.0%	1	3.8%	7
Wrong tool/action	8	20.0%	1	3.8%	9
Wrong tool argument	4	10.0%	2	7.7%	6
Verification failure	1	2.5%	4	15.4%	5
Premature abandonment	4	10.0%	1	3.8%	5
Dialogue-state failure	11	27.5%	12	46.2%	23
Unclear	1	2.5%	1	3.8%	2

The pipeline's watch items, as found here: **premature STOP before the final tool call** (the simulated user confirms and stops in one message; Figure 10); **internal ID used as a natural-language name** (the scenario's `known_info` keeps raw user IDs, which the Bangla user gives as a name); **changed task constraint** (budget or choice changed by the simulated user); **compound task partially completed** (the chat ends after the first of two requests); **early transfer** (###TRANSFER### before the agent calls the transfer tool).

7. Confound type (pipeline section 7)

Confound	gemma	share of 40	gpt-oss	share of 26
C1 Genuine Bangla-related agent failure	6	15.0%	2	7.7%
C2 Bangla user-simulator error	19	47.5%	16	61.5%
C3 Translation/localization error	1	2.5%	0	0.0%
C4 Tool/environment failure	0	0.0%	0	0.0%
C5 Benchmark/task ambiguity	0	0.0%	0	0.0%
C6 General model weakness present in both languages	14	35.0%	7	26.9%
C7 Unclear	0	0.0%	1	3.8%

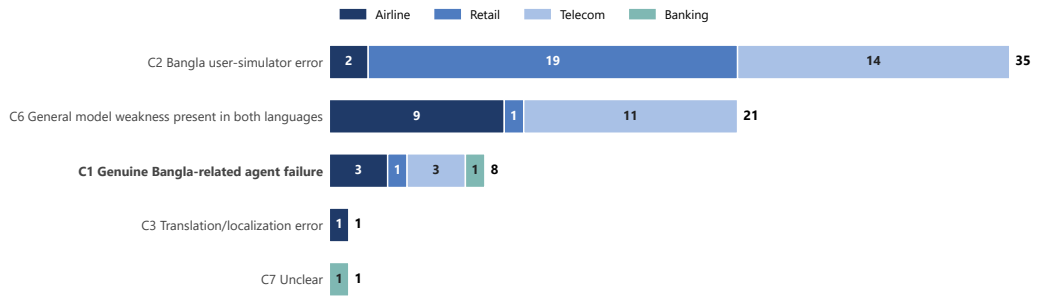


Figure 12. Confound class of the 66 EN-pass/BN-fail cases (both models), split by domain.

Confidence of the annotators: high 15, medium 44, low 7. Retail EN-pass/BN-fail cases are almost all simulator-caused; telecom mixes simulator, general and genuine causes; airline is mostly general weakness.

8. Root-cause decision tree (pipeline section 8)

Each case leaves the tree at the first step that fires. Exits are derived from the confound and the first failure point: Step 1 = translation (C3), Step 2 = user simulator (C2), otherwise by first failure point (understanding / grounding / constraint → Step 3, planning or tool selection → Step 4, tool argument → Step 5, tool execution → Step 6, verification / premature termination / final response → Step 7).

Exit step	gemma	gpt-oss
Step 1: translated task not valid	1	0
Step 2: user simulator behaved incorrectly	19	16
Step 3: agent misunderstood / mis-grounded	3	3
Step 4: wrong plan, tool or action	10	3
Step 5: wrong argument	4	2
Step 6: tool/environment	0	0
Step 7: verification / abandonment	3	1
No exit (unclear)	0	1

Step 2 is the largest exit for both models. No case exits at Step 6: the tools never failed on a valid call.

9. Evidence for each root cause (pipeline section 9)

Each of the notes carries the English outcome, the Bangla outcome, the first divergence, the failed or missing action, the confound with confidence, and an evidence quote (a user or agent message or a tool call). They are in Appendix A.

10. Counts (pipeline section 10)

Count	gemma	gpt-oss	both	share of 66
Genuine Bangla-related agent failures (C1), after filtering	6	2	8	12.1%
Bangla user-simulator-caused (C2)	19	16	35	53.0%
Translation-caused (C3)	1	0	1	1.5%
Tool/environment-caused (C4)	0	0	0	0.0%
Benchmark/task ambiguity (C5)	0	0	0	0.0%
General weakness present in both (C6)	14	7	21	31.8%
Unclear (C7)	0	1	1	1.5%

Domain-wise: Figure 12. Model-wise: gemma 40, gpt-oss 26. The upper bound on genuine Bangla agent effects is C1 + C7 = 9 cases (13.6%); the lower bound is C1 = 8 (12.1%).

11. Statistical analysis (pipeline section 11)

Paired design: the same tasks in both languages. McNemar exact two-sided test on discordant pairs; 95% intervals: Wilson for proportions, paired bootstrap (4000 resamples of tasks, seed 7) for the gap and MELR. Effect size: Cohen's g = b/(b+c) − 0.5 and the discordant odds ratio b/c. Holm across the four domains of each model, and separately across the two models' overall tests.

Model	Setting	EN [95% CI]	BN [95% CI]	Gap [95% CI]	b / c	McNemar p	Holm p	Cohen's g	Odds b/c
gemma4:31b	**Overall**	**49.5% [43.0, 56.1]**	**38.1% [31.9, 44.7]**	**+11.5 pp [5.0, 17.4]**	**40 / 15**	**0.001**	**0.002**	**0.23**	**2.67**
	Airline	66.0% [52.2, 77.6]	46.0% [33.0, 59.6]	+20.0 pp [6.0, 34.0]	12 / 2	0.013	0.052	0.36	6.00
	Retail	73.2% [58.1, 84.3]	51.2% [36.5, 65.7]	+22.0 pp [7.3, 36.6]	11 / 2	0.022	0.067	0.35	5.50
	Telecom	47.1% [37.0, 57.5]	41.4% [31.6, 51.9]	+5.7 pp [-5.7, 17.2]	16 / 11	0.442	0.884	0.09	1.45
	Banking	10.0% [4.0, 23.1]	7.5% [2.6, 19.9]	+2.5 pp [0.0, 7.5]	1 / 0	1.000	1.000	0.50	∞
gpt-oss:20b	**Overall**	**38.9% [31.5, 46.9]**	**39.6% [32.1, 47.6]**	**-0.7 pp [-10.7, 8.7]**	**26 / 27**	**1.000**	**1.000**	**-0.01**	**0.96**
	Airline	39.6% [27.0, 53.7]	45.8% [32.6, 59.7]	-6.2 pp [-18.8, 6.2]	3 / 6	0.508	1.000	-0.17	0.50
	Retail	43.9% [29.9, 59.0]	26.8% [15.7, 41.9]	+17.1 pp [0.0, 34.1]	10 / 3	0.092	0.369	0.27	3.33
	Telecom	39.1% [26.4, 53.5]	50.0% [36.1, 63.9]	-10.9 pp [-32.6, 13.0]	12 / 17	0.458	1.000	-0.09	0.71
	Banking	21.4% [7.6, 47.6]	21.4% [7.6, 47.6]	+0.0 pp [-21.4, 21.4]	1 / 1	1.000	1.000	0.00	1.00

gemma's overall gap is significant (p = 0.001, Holm 0.002); no single domain survives Holm. gpt-oss shows no gap. The model difference on the shared 149 tasks is +14.1 pp, interval [2.0, 26.2] pp. Tasks are clustered (the same simulated customer recurs across tasks), so p-values are optimistic.

Sensitivity analysis

Scenarios remove confounded tasks, or re-score runs where the grader is demonstrably wrong. **One-sided** scenarios (S1–S4) remove only EN-pass/BN-fail tasks, because the EN-fail/BN-pass side has no trace notes: they push the gap down mechanically. **Symmetric** scenarios (S5–S7) treat both languages the same way.

gemma4:31b

Scenario	Kind	n	EN	BN	Gap	Gap 95% CI (pp)	MELR	PF / FP	McNemar p
S0 As scored	symmetric	218	49.5%	38.1%	+11.5 pp	[5.0, 17.4]	23.1%	40 / 15	0.001
S1 Drop translation cases (C3)	one-sided	217	49.3%	38.2%	+11.1 pp	[4.6, 17.5]	22.4%	39 / 15	0.001
S2 Drop user-simulator cases (C2)	one-sided	199	44.7%	41.7%	+3.0 pp	[-3.0, 9.0]	6.7%	21 / 15	0.405
S3 Drop C2 + C3	one-sided	198	44.4%	41.9%	+2.5 pp	[-3.0, 8.1]	5.7%	20 / 15	0.500
S4 Drop C2 + C3 + C6 (leave C1 and C7)	one-sided	184	40.2%	45.1%	-4.9 pp	[-9.8, 0.0]	-12.2%	6 / 15	0.078
S5 Credit answers with Bengali digits (grader fix)	symmetric	218	49.5%	39.0%	+10.6 pp	[4.1, 17.0]	21.3%	39 / 16	0.003
S6 Drop premature-stop runs in either language	symmetric	156	58.3%	51.3%	+7.1 pp	[-0.0, 14.1]	12.1%	23 / 12	0.090
S7 S6 and S5 together	symmetric	156	58.3%	52.6%	+5.8 pp	[-1.9, 12.8]	9.9%	22 / 13	0.175

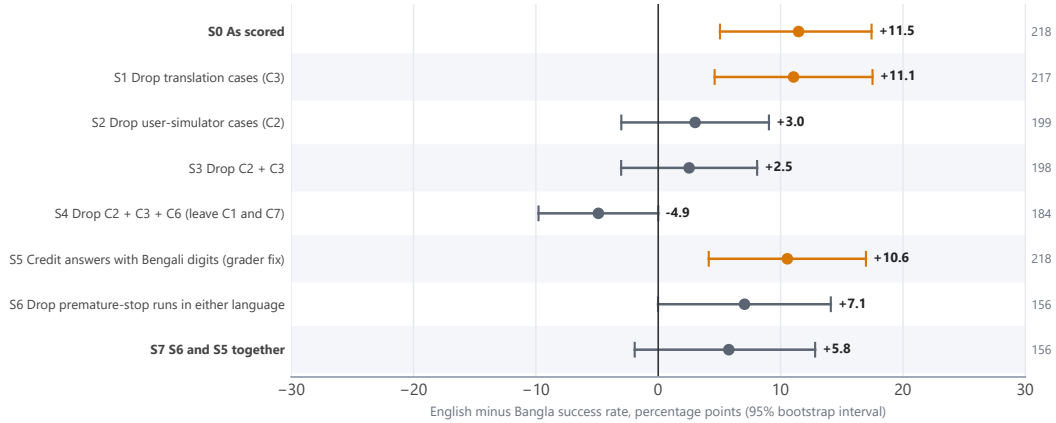


Figure 13. gemma4:31b: English minus Bangla gap with 95% interval under each sensitivity scenario. Amber intervals exclude zero.

gpt-oss:20b

Scenario	Kind	n	EN	BN	Gap	Gap 95% CI (pp)	MELR	PF / FP	McNemar p
S0 As scored	symmetric	149	38.9%	39.6%	-0.7 pp	[-10.7, 8.7]	-1.7%	26 / 27	1.000
S1 Drop translation cases (C3)	one-sided	149	38.9%	39.6%	-0.7 pp	[-10.7, 8.7]	-1.7%	26 / 27	1.000
S2 Drop user-simulator cases (C2)	one-sided	133	31.6%	44.4%	-12.8 pp	[-21.1, -4.5]	-40.5%	10 / 27	0.008
S3 Drop C2 + C3	one-sided	133	31.6%	44.4%	-12.8 pp	[-21.1, -4.5]	-40.5%	10 / 27	0.008
S4 Drop C2 + C3 + C6 (leave C1 and C7)	one-sided	126	27.8%	46.8%	-19.0 pp	[-27.0, -11.1]	-68.6%	3 / 27	<0.001
S5 Credit answers with Bengali digits (grader fix)	symmetric	149	38.9%	39.6%	-0.7 pp	[-10.7, 8.7]	-1.7%	26 / 27	1.000
S6 Drop premature-stop runs in either language	symmetric	143	40.6%	41.3%	-0.7 pp	[-10.5, 9.1]	-1.7%	26 / 27	1.000
S7 S6 and S5 together	symmetric	143	40.6%	41.3%	-0.7 pp	[-10.5, 9.1]	-1.7%	26 / 27	1.000

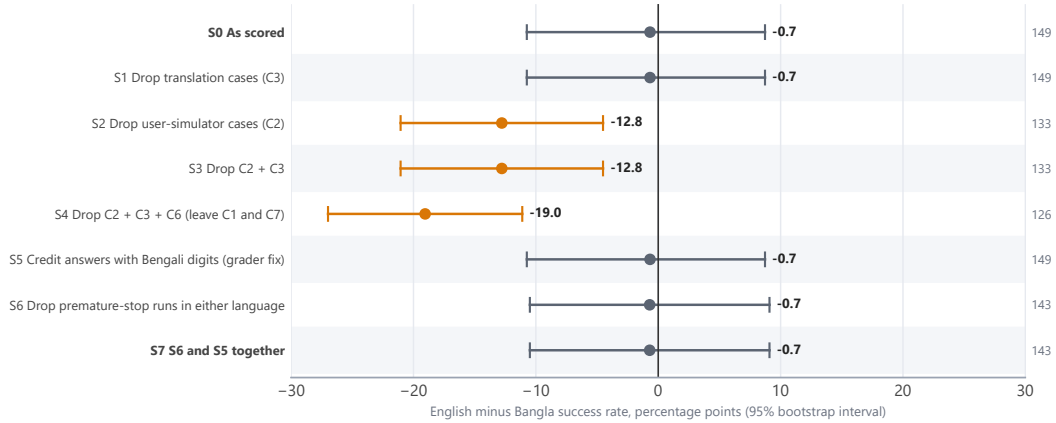


Figure 14. gpt-oss:20b: English minus Bangla gap with 95% interval under each sensitivity scenario. Amber intervals exclude zero.

How to read this. For gemma, removing the dialogues the simulated user ended early, in both languages (S6), takes the gap from +11.5 pp to +7.1 pp (p = 0.090), and crediting Bengali-digit answers as well (S7) to +5.8 pp (p = 0.175). The one-sided simulator filter (S2) gives +3.0 pp, but it is biased: applied to gpt-oss it turns no gap into a Bangla *advantage* (-12.8 pp, p = 0.008). Two cautions apply: the premature-stop detector has modest precision (a hand check of 6 random gemma flags found about 3 true premature stops), and S6 drops about a quarter of gemma's pairs.

12. Findings (pipeline section 12)

Each finding gives observation, evidence, root cause and boundary.

F1. The raw Bangla drop is model-dependent. gemma loses 11.5 pp (MELR 23.1%, p = 0.001) and the drop replicates on 219 different tasks; gpt-oss loses nothing (-0.7 pp). The difference between the models is +14.1 pp, interval [2.0, 26.2]. *Boundary:* one trial per cell; tasks share simulated customers, so p-values are optimistic.

F2. The largest driver is the Bangla user simulator. 35 of 66 EN-pass/BN-fail cases (53%) are user-simulator errors; 10 of them start at premature termination (confirm and ###STOP### in the same message, Figure 10). The automatic detector flags 46 gemma Bangla runs against 26 English runs. *Boundary:* the simulator is the same model as the agent, so a weaker Bangla simulator and a weaker Bangla agent come from the same model.

F3. Genuine Bangla agent failures are few and sit in grounding and telecom tool use. 8 cases are coded genuine: the Bangla agent sends an ID or a Bangla-script name where a lookup needs the English one, or calls device-side tools itself instead of instructing the user. Tool syntax is never the first failure point. *Boundary:* low counts, mostly medium confidence.

F4. Telecom is where the simulator matters most. The user must act on a simulated phone; the Bangla user often narrates an action ("I restarted the phone") without the device tool call, so the device check fails. gpt-oss is *better* in Bangla on telecom (-10.9 pp). *Boundary:* 19 gpt-oss telecom tasks remain unpaired.

F5. Grader artifacts are real but small. Crediting answers written with Bengali digits flips 2 gemma Bangla runs and no English run. *Boundary:* only the communicate check was re-scored; database and action checks were kept.

Not claimed. The data do not support "Bangla breaks tool calling": no case fails first on tool syntax, and gpt-oss shows no Bangla cost at all.

13. Paper outputs (pipeline section 13)

#	Output	Status	Where
1	Overall English vs Bangla success table	Done	Section R, Section 2
2	MELR by dataset and model	Done (by domain, both models, teammate set)	Section 2, Section 11
3	Paired outcome table	Done	Section 2, Figures 4–6
4	Failure-type distribution for PF	Done	Section 6
5	First-failure-point distribution	Done	Section 5
6	Confound filtering table	Done	Section 7, Section 10
7	Before/after confound-filtered results	Done (one-sided and symmetric)	Section 11
8	Two or three short case studies	Done	Section 15
9	Appendix with annotation rules and long trajectories	Notes done; rules in `docs/tau2/annotation_rules_and_cases.md`	Appendix A

14. Quality-control checklist (pipeline section 14)

Check	Status	Note
Every task ID aligned between English and Bangla	Pass	Keyed by (model, domain, task ID); infrastructure-error pairs dropped on both sides.
Every metric has a denominator	Pass	Stated in each table.
EN-pass/BN-fail separated from both-fail	Pass	Sections 2, 3; Appendix B.
Translation errors not counted as genuine	Pass	C3 is its own class.
User-simulator errors counted separately	Pass	C2 is its own class (35 cases).
Tool/environment failures counted separately	Pass	C4 is its own class (0 cases).
A second reviewer checks a subset	Partly	46 earlier notes: second review and adjudication (κ = 0.39 fine confound, 0.71 artifact vs agent). 20 newest notes: **not done**.
Disagreements adjudicated	Partly	Earlier 46 yes; newest 20 no.
Case studies match the quantitative patterns	Pass	Section 15 cases come from C2 (largest class), C1 and C6.
Strong claims only after confound filtering	Pass	F1–F5 state boundaries; one-sided filters are labelled as biased.

Blind subset for a second reviewer of the newest notes (random, seed 11): `telecom [mms_issue]bad_network_preference|user_abroad_ro… (gemma)`, `telecom [mobile_data_issue]airplane_mode_on|bad_network_… (gemma)`, `telecom [mobile_data_issue]bad_network_preference|bad_vp… (oss)`, `telecom [mobile_data_issue]airplane_mode_on|bad_network_… (oss)`, `telecom [service_issue]airplane_mode_on|break_apn_settin… (gemma)`, `telecom [service_issue]overdue_bill_suspension|unseat_si… (oss)`, `telecom [service_issue]airplane_mode_on|break_apn_settin… (gemma)`, `telecom [mobile_data_issue]bad_network_preference|bad_vp… (gemma)`.

15. Case studies

airline 19 (gemma4:31b): A correct agent, cut off by the simulated user. *English:* Agent refused to modify basic economy, user asked to cancel with insurance, agent called cancel_reservation after confirmation; passed. *Bangla:* Agent asked to confirm cancelling Z7GOZK; the Bangla user confirmed and appended ###STOP### in the same message, ending the dialogue before cancel_reservation was called.

Root cause: Final BN user turn combined confirmation with ###STOP###, so the agent never got to execute the cancellation. *Label:* C2, Bangla user-simulator error, first failure point "Premature termination".

airline 9 (gemma4:31b): A genuine Bangla agent failure. *English:* Agent noticed NQNU5R flights (May 13-14) already flown, refused cancel, searched JFK-MCO, and transferred; reward 1. *Bangla:* Agent claimed NQNU5R can be cancelled since business class, got the user to confirm, and called cancel_reservation on an already-flown reservation.

Root cause: BN agent failed to check the flight dates against the policy (flown flights cannot be cancelled), unlike EN which flagged it. *Label:* C1, Genuine Bangla-related agent failure, first failure point "Planning".

airline 8 (gemma4:31b): A failure the model also makes in English. *English:* Agent looked up the profile and past flight, booked HAT271 economy for 2 passengers with 0 bags, certificate_8045380; DB matched. *Bangla:* Agent found HAT271 and booked the same flight/passengers/payment, but invented '2 free bags' and passed total_baggages=2 although the user wanted no baggage.

Root cause: In the BN confirmation summary the agent claimed 'ব্যাগ: মোট ২ টি (free)' and then booked with total_baggages=2, whereas EN used 0. *Label:* C6, General model weakness (not Bangla-specific), first failure point "Tool-argument construction".

Appendix A. Trace notes for all 66 EN-pass/BN-fail cases

Grouped by model, ordered by domain and task ID. Fields follow pipeline section 4; confound with annotator confidence.

gemma4:31b (40 cases)

airline 3 · gemma4:31b · C3

- **Task goal:** You want to figure out the total number of suitcases the reservation allows you to take on your upcoming flight. You have a lot of things you need to bring with you on this trip. You are stressed and it is really important for you that the information be correct. You're pretty sure that you're a Gold member.
- **English:** Agent stated 4 free bags in Western digits, user pushed on Gold, agent held Silver and transferred; communicate check '4' met.
- **Bangla:** Agent gave the same answer but wrote it as Bengali digits (৪টি ফ্রি ব্যাগ), then user transferred; the string-match communicate check '4' failed while DB matched.
- **First divergence:** Same trajectory; BN diverged only in numeral script (Bengali digit vs '4'); the BN user also did not ask for numeric form.
- **Final failed / missing action:** none (agent communicated 4 bags as '৪'; evaluator expects ASCII '4')
- **First failure point:** Final response only · **Type:** Unclear
- **Confound:** Translation/localization error (high)
- **Evidence:** "আপনারা মোট **৪টি ফ্রি ব্যাগ** নিতে পারবেন"; COMMUNICATE: Information '4' not communicated

airline 8 · gemma4:31b · C6

- **Task goal:** You want to book a one-way flight from ORD to PHL on May 26.
- **English:** Agent looked up the profile and past flight, booked HAT271 economy for 2 passengers with 0 bags, certificate_8045380; DB matched.
- **Bangla:** Agent found HAT271 and booked the same flight/passengers/payment, but invented '2 free bags' and passed total_baggages=2 although the user wanted no baggage.
- **First divergence:** In the BN confirmation summary the agent claimed 'ব্যাগ: মোট ২ টি (free)' and then booked with total_baggages=2, whereas EN used 0.
- **Final failed / missing action:** book_reservation total_baggages should be 0 (got 2)
- **First failure point:** Tool-argument construction · **Type:** Wrong tool argument
- **Confound:** General model weakness present in both languages (high)
- **Evidence:** book_reservation(... "total_baggages": 2 ...) after user said 'আমার কোনো অতিরিক্ত ব্যাগের প্রয়োজন নেই'

airline 9 · gemma4:31b · C1

- **Task goal:** You want to cancel two of your upcoming reservations (IFOYYZ and NQNU5R) and change a third (M20IZO) to a nonstop flight if available.
- **English:** Agent noticed NQNU5R flights (May 13-14) already flown, refused cancel, searched JFK-MCO, and transferred; reward 1.
- **Bangla:** Agent claimed NQNU5R can be cancelled since business class, got the user to confirm, and called cancel_reservation on an already-flown reservation.
- **First divergence:** BN agent failed to check the flight dates against the policy (flown flights cannot be cancelled), unlike EN which flagged it.
- **Final failed / missing action:** Must NOT cancel NQNU5R (flights already departed); BN called cancel_reservation(NQNU5R)
- **First failure point:** Planning · **Type:** Wrong tool/action
- **Confound:** Genuine Bangla-related agent failure (high)
- **Evidence:** cancel_reservation({"reservation_id": "NQNU5R"}) after "এটি একটি বিজনেস ক্লাস ফ্লাইট, তাই এটি বাতিল করা সম্ভব"

airline 12 · gemma4:31b · C2

- **Task goal:** You have an upcoming flight from Boston to Minneapolis under reservation ID YAX4DR. You want to change your class for all passengers to business. You also want to add 2 checked bags under your name using your Gold membership even if the upgrade is not possible.
- **English:** Agent declined single-passenger upgrade, then added 2 bags via update_reservation_baggages after user confirmation; passed.
- **Bangla:** Agent asked to confirm adding 2 bags; the Bangla user said yes and appended ###STOP### in the same message, so the conversation ended before update_reservation_baggages was called.
- **First divergence:** Final BN user turn combined confirmation with ###STOP###, ending the dialogue before the agent's required write.
- **Final failed / missing action:** update_reservation_baggages(YAX4DR, total_baggages=2, nonfree=0) never executed
- **First failure point:** Premature termination · **Type:** Dialogue-state failure
- **Confound:** Bangla user-simulator error (high)
- **Evidence:** [USER] হ্যাঁ, দয়া করে আমার জন্য ২টি checked bag যোগ করে দিন। ধন্যবাদ। ###STOP###

airline 16 · gemma4:31b · C6

- **Task goal:** For your upcoming trip from ATL to PHL, you want to change for the cheapest flight in Economy cabin class (not Basic Economy) and for the day after the original reservation.
- **English:** Agent searched ATL->PHL direct and one-stop for 05-24 and picked cheapest economy via LGA ($207); passed.
- **Bangla:** Agent anchored on the original itinerary legs, searched ATL->ORD and ORD->PHL separately, and booked HAT227+HAT139 economy ($216), never searching ATL->PHL one-stop options.
- **First divergence:** BN agent's searches (ATL->ORD, ORD->PHL) instead of ATL->PHL direct/one-stop, so it missed the cheaper HAT110+HAT172 route.
- **Final failed / missing action:** update_reservation_flights with cheapest economy HAT110/HAT172 (got HAT227/HAT139)
- **First failure point:** Tool-argument construction · **Type:** Wrong tool argument
- **Confound:** General model weakness present in both languages (medium)
- **Evidence:** search_direct_flight({date:2024-05-24,destination:ORD,origin:ATL}) ... update_reservation_flights HAT227(112)+HAT139(104)

airline 19 · gemma4:31b · C2

- **Task goal:** You will have a crazy half-day trip to Texas. It is in your reservations but you don't remember the reservation id. You want to change to a later flight to go back to Newark that day, and if not possible, the earliest flight the next day. Your current return flight departs 3pm (on the 28th).
- **English:** Agent refused to modify basic economy, user asked to cancel with insurance, agent called cancel_reservation after confirmation; passed.
- **Bangla:** Agent asked to confirm cancelling Z7GOZK; the Bangla user confirmed and appended ###STOP### in the same message, ending the dialogue before cancel_reservation was called.
- **First divergence:** Final BN user turn combined confirmation with ###STOP###, so the agent never got to execute the cancellation.
- **Final failed / missing action:** cancel_reservation(Z7GOZK) never executed
- **First failure point:** Premature termination · **Type:** Dialogue-state failure
- **Confound:** Bangla user-simulator error (high)
- **Evidence:** [USER] হ্যাঁ, আমি নিশ্চিত। দয়া করে আমার রিজার্ভেশন Z7GOZK বাতিল করে দিন। ###STOP###

airline 22 · gemma4:31b · C6

- **Task goal:** For your upcoming trip from New York to Chicago, you want to change the passenger to yourself, upgrade it to economy class, and have 3 checked bags.
- **English:** Agent asked which payment method to use, user chose gift_card_8190333, and all three updates succeeded with that card.
- **Bangla:** Agent never asked for payment method and silently used credit_card_6754990 for the flights and baggage updates; the user (told not to volunteer unasked info) never mentioned the gift card.
- **First divergence:** After the user's Bangla confirmation, BN skipped the payment-method question that EN asked and picked a card on its own.
- **Final failed / missing action:** update_reservation_flights and update_reservation_baggages with payment_id gift_card_8190333 (BN used credit_card_6754990)
- **First failure point:** Tool-argument construction · **Type:** Wrong tool argument
- **Confound:** General model weakness present in both languages (high)
- **Evidence:** update_reservation_flights({... "payment_id": "credit_card_6754990" ...}); action_checks 22_0, 22_2 false, 22_1 true

airline 26 · gemma4:31b · C6

- **Task goal:** You want to cancel your flights from MCO to CLT.
- **English:** Agent correctly applied policy (basic economy, no insurance, >24h) and refused, offering transfer; user asked for transfer.
- **Bangla:** Agent hallucinated that cancellation is allowed because the user is a gold member, got confirmation, and cancelled the basic-economy reservation.
- **First divergence:** BN agent asserted an invented policy ('possible because you are gold') instead of checking cancellation criteria as EN did.
- **Final failed / missing action:** Must refuse cancellation (basic economy, no insurance, booked >24h); BN called cancel_reservation(3FRNFB)
- **First failure point:** Planning · **Type:** Planning failure
- **Confound:** General model weakness present in both languages (medium)
- **Evidence:** "নীতি অনুযায়ী, এই রিজার্ভেশনটি বাতিল করা সম্ভব কারণ আপনি একজন gold মেম্বার"; cancel_reservation(3FRNFB)

airline 34 · gemma4:31b · C1

- **Task goal:** You want to change your upcoming outgoing flight in reservation HXDUBJ to a nonstop flight on the next day (i.e. delay by one day). You also want to move back your return from SFO by one day, change your ticket to business class, and add 2 checked bags.
- **English:** Agent offered options for both legs, user picked HAT072 and HAT278 (16:00), cost computed as 222 which exceeded the $200 budget so user cancelled; no changes.
- **Bangla:** Agent offered only outbound options at economy price, then unilaterally chose HAT082 (23:00, violating user's 8am-9pm window) for return, computed 102 (under budget), and executed the changes.
- **First divergence:** BN agent never presented return-flight options and picked HAT082 itself, so the true cost (222) and budget trigger never surfaced.
- **Final failed / missing action:** Should have made no changes (package cost 222 > $200 budget); BN executed update_reservation_flights and baggages
- **First failure point:** Entity or slot grounding · **Type:** Planning failure
- **Confound:** Genuine Bangla-related agent failure (medium)
- **Evidence:** calculate("282 + 323 - (177 + 146 + 180)") = 102; update_reservation_flights ... HAT082 payment gift_card_6941833

airline 40 · gemma4:31b · C6

- **Task goal:** You booked the flight and you want to change the passenger name on the reservation.
- **English:** Agent reused Mei's DOB from the reservation, confirmed, and called update_reservation_passengers with Mei Garcia.
- **Bangla:** Agent asked the user for Mei's date of birth (already present in the reservation), user said he did not have it, agent refused to proceed and transferred to a human.
- **First divergence:** At the name-change step BN agent requested a DOB instead of copying it from get_reservation_details output.
- **Final failed / missing action:** Missing update_reservation_passengers(3RK2T9, [Anya Garcia 1992-11-12, Mei Garcia 1989-12-13]).
- **First failure point:** Entity or slot grounding · **Type:** Planning failure
- **Confound:** General model weakness present in both languages (medium)
- **Evidence:** 'সঠিক তথ্যের জন্য অনুগ্রহ করে Mei Garcia-এর জন্ম তারিখটি প্রদান করুন।' ... 'জন্ম তারিখ ছাড়া আমি নাম পরিবর্তন করতে পারছি না' -> transfer

airline 41 · gemma4:31b · C6

- **Task goal:** You want to cancel all of your upcoming flights that only have one passenger on the reservation.
- **English:** Agent listed single-passenger reservations, then re-checked policy, found UDMOP1 ineligible, refused and offered transfer; no cancellation.
- **Bangla:** Agent listed the same reservations, asked reason, then without checking cancellation eligibility asked for confirmation and cancelled basic-economy, uninsured UDMOP1.
- **First divergence:** After the user gave 'change of plan', EN agent re-verified policy and refused; BN agent skipped the eligibility check and asked for confirmation.
- **Final failed / missing action:** Agent must not cancel any reservation; cancel_reservation(UDMOP1) was wrongly executed.
- **First failure point:** Planning · **Type:** Verification failure
- **Confound:** General model weakness present in both languages (medium)
- **Evidence:** cancel_reservation({"reservation_id": "UDMOP1"}) after only 'আমি আপনার UDMOP1 রিজার্ভেশনটি বাতিল করার অনুরোধটি গ্রহণ করেছি' (no policy check)

airline 43 · gemma4:31b · C6

- **Task goal:** You are contacting customer support because you have booked two flights for the same day.
- **English:** Agent identified D1EW9B as ineligible (basic economy, no insurance, >24h) and transferred to a human; no cancellation.
- **Bangla:** Agent correctly refused D1EW9B, but after the user (per instructions) offered the other May 17 flight it cancelled 9HBUV8, citing a non-existent silver-member exception.
- **First divergence:** After refusing D1EW9B the EN agent transferred to a human immediately, whereas the BN agent stayed in the conversation, was pushed by the user, and then approved a second ineligible cancellation.
- **Final failed / missing action:** Should not call cancel_reservation(9HBUV8) (basic economy, insurance does not cover change of plan); it did.
- **First failure point:** Planning · **Type:** Wrong tool/action
- **Confound:** General model weakness present in both languages (medium)
- **Evidence:** 'পরিকল্পনার পরিবর্তন ইন্স্যুরেন্স দ্বারা কভার করা হয় না। তবে, যেহেতু আপনি একজন silver member, তাই আমি এটি বাতিল করতে পারি।' -> cancel_reservation(9HBUV8)

banking task_036 · gemma4:31b · C1

- **Task goal:** **Your character:** You are Fatima Al-Hassan, a 31-year-old small business owner from Detroit, MI. You have a Silver Rewards Card with Rho-Bank. **Your situation:** You just checked your credit card statement and something feels off - your balance is higher than you expected. You want to see your recent transactions to figure out what's going on. You're calling Rho-Bank to get an overview of your last several transac…
- **English:** After verification and showing transactions, user reported fraud; agent ran KB_search, unlocked order_replacement_credit_card_7291 and ordered an expedited replacement to the work address; DB matched.
- **Bangla:** Verification and transactions identical (agent used get_user_information_by_name and logged verification correctly). When the user asked to cancel the card and get a new one, agent did not search the KB or look for discoverable tools, claimed it had no ability to cancel/replace, and offered a human transfer; user accepted and the run ended in ###TRANSFER###.
- **First divergence:** BN agent turn #9-#10 (after user reports fraud): offered transfer to a human agent instead of KB_search / unlock_discoverable_agent_tool as EN did.
- **Final failed / missing action:** Never called unlock_discoverable_agent_tool(order_replacement_credit_card_7291) nor call_discoverable_agent_tool with fraud_suspected, work address, expedited_shipping=true.
- **First failure point:** Tool selection · **Type:** Wrong tool/action
- **Confound:** Genuine Bangla-related agent failure (medium)
- **Evidence:** "কার্ড বাতিল করা বা নতুন কার্ড ইস্যু করার মতো কাজগুলো সরাসরি আমার করার ক্ষমতা নেই"

retail 0 · gemma4:31b · C2

- **Task goal:** You received your order #W2378156 and wish to exchange the mechanical keyboard for the same one but with clicky switches and the smart thermostat for one compatible with Google Home instead of Apple HomeKit. If there is no keyboard that is clicky, RGB backlight, full size, you'd go for no backlight.
- **English:** Agent identified Yusuf Rossi, exchanged keyboard (7706410293) and thermostat (7747408585) on credit_card_9513926 after a user "Yes".
- **Bangla:** Agent proposed the same correct exchange, but the user simulator confirmed and emitted ###STOP### in the same message, so exchange_delivered_order_items was never called.
- **First divergence:** At the confirmation turn the BN user wrote the "yes" and ###STOP### together, ending the episode before the write call (EN user gave a bare "Yes").
- **Final failed / missing action:** exchange_delivered_order_items (action 0_4) never called; db_match false.
- **First failure point:** Premature termination · **Type:** Dialogue-state failure
- **Confound:** Bangla user-simulator error (high)
- **Evidence:** USER: "...অনুগ্রহ করে প্রসেসটি শুরু করুন। ###STOP###" (agent also miscomputed refund as 26.63 vs 16.63 and used Bengali digits, but secondary)

retail 1 · gemma4:31b · C2

- **Task goal:** You received your order #W2378156 and wish to exchange the mechanical keyboard for the same one but with clicky switches and the smart thermostat for one compatible with Google Home instead of Apple HomeKit. If there is no keyboard that is clicky, RGB backlight, full size, you'd rather only exchange the thermostat.
- **English:** User confirmed thermostat-only exchange; agent then called exchange_delivered_order_items (4983901480->7747408585) and user stopped after (reward 1).
- **Bangla:** Agent reached the identical correct plan and summary, but BN user replied 'হ্যাঁ, আমি নিশ্চিত করছি...' with ###STOP### in the same message, so the episode ended before the agent made the exchange call.
- **First divergence:** Final confirmation turn: user simulator emitted ###STOP### together with the confirmation (artifact pattern a); the agent had no turn to act.
- **Final failed / missing action:** exchange_delivered_order_items(#W2378156,[4983901480],[7747408585],credit_card_9513926) never called
- **First failure point:** Premature termination · **Type:** Dialogue-state failure
- **Confound:** Bangla user-simulator error (high)
- **Evidence:** USER: 'হ্যাঁ, আমি নিশ্চিত করছি। অনুগ্রহ করে এক্সচেঞ্জটি সম্পন্ন করুন। ###STOP###'

retail 5 · gemma4:31b · C2

- **Task goal:** You want to exchange the water bottle and the desk lamp. You want to exchange the water bottle to a bigger one, and the desk lamp to a less bright one (prefer battery > USB > AC). If the agent asks for confirmation, only exchange the desk lamp. If the agent asks for confirmation again, do not exchange anything, and return the water bottle instead.
- **English:** Agent authenticated as Mei Kovacs, offered exchanges; user changed mind to lamp, then to returning the water bottle; return executed (reward 1).
- **Bangla:** BN user gave 'mei_kovacs_8020' as name; agent called find_user with first_name=id, last_name empty, failed repeatedly and transferred to human.
- **First divergence:** BN user simulator stated the user-id as its name at the first auth turn; agent treated it as a name, never used the id with get_user_details.
- **Final failed / missing action:** authentication, order lookup and return_delivered_order_items(#W6390527,[8538875209]) all missing
- **First failure point:** Entity or slot grounding · **Type:** Understanding failure
- **Confound:** Bangla user-simulator error (medium)
- **Evidence:** find_user_id_by_name_zip({first_name:'mei_kovacs_8020', last_name:''}) -> User not found; transfer_to_human_agents

retail 7 · gemma4:31b · C2

- **Task goal:** You want to exchange the water bottle and the desk lamp. You want to exchange the water bottle to a bigger one, and the desk lamp to a less bright one (prefer AC adapter > battery > USB). If the agent asks for confirmation, only exchange the desk lamp.
- **English:** User gave "Mei Kovacs" and zip, agent authenticated, then exchanged only the desk lamp after the user changed mind (reward 1).
- **Bangla:** BN user gave the user_id "mei_kovacs_8020" as their name; agent searched first_name "mei", last_name "kovacs_8020", got not found, and transferred to a human after 3 turns.
- **First divergence:** First user turn after auth request: BN simulator wrote name as mei_kovacs_8020 (id copied from known_info) instead of Mei Kovacs; agent never tried splitting it.
- **Final failed / missing action:** find_user_id_by_name_zip(Mei, Kovacs, 28236), get_user_details, order lookup, product lookups and exchange_delivered_order_items(#W6390527,[8384507844]->[1569765161]) all missing
- **First failure point:** Entity or slot grounding · **Type:** Understanding failure
- **Confound:** Bangla user-simulator error (medium)
- **Evidence:** find_user_id_by_name_zip({"first_name":"mei","last_name":"kovacs_8020","zip":"28236"}) -> User not found; user: "আমার নাম mei_kovacs_8020"

retail 8 · gemma4:31b · C2

- **Task goal:** You want to exchange the water bottle and the desk lamp. You want to exchange the water bottle to a bigger one, and the desk lamp to a brighter one (prefer battery > USB > AC). If the agent asks for confirmation, only exchange the desk lamp.
- **English:** Agent authenticated as Mei Kovacs, offered options, user dropped the water bottle, and the desk-lamp-only exchange was executed (reward 1).
- **Bangla:** BN user gave the user-id 'mei_kovacs_8020' as their name; agent split it into first_name 'mei' / last_name 'kovacs_8020', got User not found 3 times and transferred to a human; no exchange made.
- **First divergence:** At the authentication turn the BN user simulator supplied the id string as 'my name' instead of 'Mei Kovacs'; agent never tried get_user_details with the id or the proper name.
- **Final failed / missing action:** find_user_id_by_name_zip with Mei/Kovacs/28236 and all downstream reads plus exchange_delivered_order_items missing; transfer_to_human_agents used instead
- **First failure point:** Entity or slot grounding · **Type:** Understanding failure
- **Confound:** Bangla user-simulator error (medium)
- **Evidence:** USER: 'আমার নাম mei_kovacs_8020 ...' -> find_user_id_by_name_zip({first_name:'mei',last_name:'kovacs_8020'}) -> Error: User not found -> transfer_to_human_agents

retail 9 · gemma4:31b · C2

- **Task goal:** You want to exchange the water bottle and the desk lamp. You want to exchange the water bottle to a bigger one, and the desk lamp to a brighter one (prefer AC adapter > battery > USB). When the agent asks for confirmation, suddenly change your mind and ask to only exchange the desk lamp.
- **English:** User gave "Mei Kovacs" and zip, agent authenticated, found order, and exchanged only the desk lamp to 7624783998 (reward 1).
- **Bangla:** BN user gave the user_id "mei_kovacs_8020" as the name; agent searched first "mei", last "kovacs_8020" twice, failed, and transferred to a human.
- **First divergence:** Authentication turn: BN simulator wrote the id as the name; agent never tried "Mei Kovacs" or other split.
- **Final failed / missing action:** Authentication and the exchange_delivered_order_items(#W6390527,[8384507844]->[7624783998]) never happened
- **First failure point:** Entity or slot grounding · **Type:** Understanding failure
- **Confound:** Bangla user-simulator error (medium)
- **Evidence:** find_user_id_by_name_zip({"first_name":"mei","last_name":"kovacs_8020"...}) -> "Error: User not found"; user: "আমার নাম mei_kovacs_8020"

retail 10 · gemma4:31b · C2

- **Task goal:** For some reason, you want to return all things ordered. You have two payment methods and two orders, and you want to refund each order to the other order's payment method. If not possible, you are angry and swear for a few times, then ask for human representative. Make sure to return BOTH orders.
- **English:** User wanted both orders returned with cross-refunds to the other payment method; agent refused per policy and, on user frustration, transferred to a human (matching expected action).
- **Bangla:** BN user stated the refund mapping as the original payment methods (W5490111->credit card, W7387996->PayPal), losing the "other method" goal; agent complied and returned items, never transferring.
- **First divergence:** At the refund-method turn the BN user switched from cross-refund to original-method refunds (EN user asked for swapped methods).
- **Final failed / missing action:** Missing transfer_to_human_agents; instead return_delivered_order_items called for both orders.
- **First failure point:** Task constraint preservation · **Type:** Dialogue-state failure
- **Confound:** Bangla user-simulator error (high)
- **Evidence:** USER: "অর্ডার #W5490111 এর রিফান্ড যেন Credit Card-এ যায় এবং #W7387996 PayPal-এ" (these are the originals)

retail 14 · gemma4:31b · C2

- **Task goal:** You just quit gaming and want to cancel or return everything associated with it. (It's just a keyboard and a mouse, but do not reveal it to the agent). Original payment is preferred.
- **English:** Agent inspected both orders, identified keyboard and gaming mouse as gaming-related, and returned both to original payment methods.
- **Bangla:** BN user invented a nonexistent order ID, then asked to return only order #W5490111 keyboard; agent never inspected #W7387996 so the Gaming Mouse return was missed.
- **First divergence:** After the order list, the BN user picked only one order (EN agent proactively inspected both orders and found the mouse).
- **Final failed / missing action:** get_order_details(#W7387996) and return_delivered_order_items for item 5796612084 (paypal_9497703) absent.
- **First failure point:** Task constraint preservation · **Type:** Dialogue-state failure
- **Confound:** Bangla user-simulator error (medium)
- **Evidence:** USER: "আমি #W5490111 অর্ডারটি রিটার্ন করতে চাই।" despite goal "gaming related সব কিছু"; agent also did not check the second order

retail 23 · gemma4:31b · C2

- **Task goal:** You want to exchange the helmet for a medium sized, red, high ventilation type, and you want to exchange the luggage set (in another order) to a two-piece black one with soft material. Lastly, you want to modify the grill you just ordered to the same type as the one you already received.
- **English:** Agent looked up products, summarised three changes, user said yes, and agent made both exchanges and the grill modify call (reward 1).
- **Bangla:** Agent summarised the changes (with item IDs it never looked up: 1008292230/1008292231, no get_product_details calls), user replied "হ্যাঁ, আমি একমত" plus ###STOP### in the same message, so no write call was made.
- **First divergence:** After the summary, the BN user emitted confirmation and ###STOP### together; agent also skipped get_product_details and fabricated item IDs.
- **Final failed / missing action:** exchange_delivered_order_items x2 (#W3947049, #W6876713) and modify_pending_order_items (#W3561391) never called; db_match false
- **First failure point:** Premature termination · **Type:** Dialogue-state failure
- **Confound:** Bangla user-simulator error (medium)
- **Evidence:** User: "হ্যাঁ, আমি একমত। আপনি এই পরিবর্তনগুলো করে দিন। ধন্যবাদ! ###STOP###"; agent text had fake item IDs 1008292230

retail 32 · gemma4:31b · C2

- **Task goal:** You just lost your tablet you just received and are in a bad mood. You want to know the tracking number of the order, and if the agent can help you refund or reorder the tablet. (You know it's a long shot, but you want to try). If not, cancel the charger you just bought, because it goes with the tablet... Also cancel the boot and kettle, and return the sneaker.
- **English:** Agent gave tracking number, cancelled two pending orders, then after user confirmed the sneaker return, called return_delivered_order_items (reward 1).
- **Bangla:** Agent gave tracking number (communicate check met), cancelled both orders correctly, then at the final return confirmation the user said "হ্যাঁ, আমি নিশ্চিত।" with ###STOP### so the return call was never made.
- **First divergence:** Last turn: BN user emitted confirmation and ###STOP### together, so the agent never executed the final return.
- **Final failed / missing action:** return_delivered_order_items(#W7449508,[6477915553],gift_card_7711863) missing (action 32_12)
- **First failure point:** Premature termination · **Type:** Dialogue-state failure
- **Confound:** Bangla user-simulator error (high)
- **Evidence:** User: "হ্যাঁ, আমি নিশ্চিত। ###STOP###" right after agent asked "আপনি কি এই পণ্যটি ফেরত দিতে নিশ্চিত?"

retail 42 · gemma4:31b · C2

- **Task goal:** You just created your user id mei_patel_7272 and ordered some things, but realized you might have typed your address wrong. You want to check it, and potentially correct all order addresses and your user address. After this, you'd like to check the jigsaw you ordered, and if it's not shipped yet, you want to change it to the easiest jigsaw (easiest level, least pieces) because your kid is too young. By default you us...
- **English:** Agent fixed addresses on user and both orders, then modified the jigsaw to item 1096508426 via PayPal after confirmation (reward 1).
- **Bangla:** Address fixes done correctly (3 write calls matched), but when the agent asked for confirmation of the jigsaw change the user confirmed and emitted ###STOP### in the same message, so modify_pending_order_items was never called.
- **First divergence:** Final jigsaw confirmation turn: BN user confirmed and ended the conversation in one message.
- **Final failed / missing action:** modify_pending_order_items(#W4082615,[9779102705]->[1096508426],paypal_4768213) missing (action 42_9)
- **First failure point:** Premature termination · **Type:** Dialogue-state failure
- **Confound:** Bangla user-simulator error (high)
- **Evidence:** User: "হ্যাঁ, এই পাজলটিই হবে। আর রিফান্ডের জন্য আমার PayPal ... ধন্যবাদ। ###STOP###"

telecom [mobile_data_issue]airplane_mode_on|data_saver_mode_on|user_abroad_roaming_disabled_on[PERSONA:None] · gemma4:31b · C6

- **Task goal:** You mobile data is not working properly. It either stops working or is very slow. You want to fix it and absolutely want to get excellent internet speed on your phone. You are not willing to accept any other internet speed (poor, fair or good). You do not have access to wifi.
- **English:** Agent identified line L1002 from the phone number, learned the user was abroad with roaming off, guided airplane/data-saver off and enabled roaming; speed became excellent (reward 1.0).
- **Bangla:** Agent never resolved which line the number 555-123-2002 belonged to, queried data usage for L1001, never learned the user was abroad nor enabled roaming on L1002, rebooted, and transferred to a human after speed tests kept failing.
- **First divergence:** After the account lookup the EN agent inspected the lines (L1002), asked about travel and found roaming disabled; the BN agent skipped line lookup and went straight to device checks and get_data_usage on L1001.
- **Final failed / missing action:** enable_roaming(C1001, L1002) never called (action_match false); end-state mobile data/speed assertions failed
- **First failure point:** Entity or slot grounding · **Type:** Understanding failure
- **Confound:** General model weakness present in both languages (medium)
- **Evidence:** get_data_usage({"customer_id": "C1001", "line_id": "L1001"}) ... transfer_to_human_agents("No Connection ... despite active 5G")

telecom [mobile_data_issue]airplane_mode_on|bad_network_preference[PERSONA:Hard] · gemma4:31b · C6

- **Task goal:** You mobile data is not working properly. It either stops working or is very slow. You want to fix it and absolutely want to get excellent internet speed on your phone. You are not willing to accept any other internet speed (poor, fair or good). You do not have access to wifi.
- **English:** Agent asked the user to run a speed test and check the status bar; user turned airplane mode off and set 4g_5g_preferred as guided; speed excellent (reward 1.0).
- **Bangla:** Agent tried to call device tools itself (run_speed_test, check_status_bar, check_network_status, check_network_mode_preference), all failed, then gave vague multi-step instructions to a fearful persona user, never had them toggle airplane mode or set network mode, and transferred to a human.
- **First divergence:** After the account lookup the BN agent called run_speed_test itself instead of asking the user to run it.
- **Final failed / missing action:** user toggle_airplane_mode and set_network_mode_preference(4g_5g_preferred) never performed; env assertions failed
- **First failure point:** Tool selection · **Type:** Wrong tool/action
- **Confound:** General model weakness present in both languages (medium)
- **Evidence:** [ACT #5] run_speed_test({}) -> Error: Tool not found; later transfer_to_human_agents(... "Direct tool access to his device failed.")

telecom [mobile_data_issue]data_saver_mode_on|data_usage_exceeded[PERSONA:Easy] · gemma4:31b · C6

- **Task goal:** You mobile data is not working properly. It either stops working or is very slow. You want to fix it and absolutely want to get excellent internet speed on your phone. You are not willing to accept any other internet speed (poor, fair or good). You do not have access to wifi.
- **English:** Agent refueled 2 GB, then guided the user through checking settings; user found Data Saver ON and turned it off, speed reached Excellent (200+).
- **Bangla:** Agent refueled 2 GB but, after its own non-existent tool calls failed (run_speed_test, check_network_status, check_network_mode_preference, check_data_restriction_status), never told the user to check/turn off Data Saver and transferred to a human; toggle_data_saver_mode never done, speed stays 'good'.
- **First divergence:** After the refuel the BN agent asked permission to check settings itself (calling non-existent tools) instead of instructing the user to inspect network mode/Data Saver/VPN as EN did, then gave up via transfer.
- **Final failed / missing action:** User action toggle_data_saver_mode missing; assert_internet_speed (excellent/200) failed; refuel was correct.
- **First failure point:** Premature termination · **Type:** Premature abandonment
- **Confound:** General model weakness present in both languages (medium)
- **Evidence:** [ACT #19] check_data_restriction_status({}) -> not found; then 'আমি কি আপনাকে একজন হিউম্যান এজেন্টের কাছে ট্রান্সফার করব?' ###TRANSFER###

telecom [mobile_data_issue]data_mode_off|data_usage_exceeded[PERSONA:None] · gemma4:31b · C2

- **Task goal:** You mobile data is not working properly. It either stops working or is very slow. You want to fix it and absolutely want to get excellent internet speed on your phone. You are not willing to accept any other internet speed (poor, fair or good). You do not have access to wifi.
- **English:** Agent looked up lines, found 15.1/15 GB exceeded, refueled 2 GB, then guided user to turn mobile data on; speed Excellent.
- **Bangla:** Agent never called get_details_by_id/refuel_data: it only asked permission to run speed test/checks on the phone, and when it asked to check data usage the user simulator replied the agent 'cannot use a tool to check my data usage' (invented/role-confused), after which the agent went to APN/reboot and transferred.
- **First divergence:** After identifying the customer the BN agent asked to run a speed test instead of inspecting the line details/plan (as EN did), and the user simulator then refused the usage check, steering away from the data-refuel fix.
- **Final failed / missing action:** refuel_data(C1001, L1002, 2.0) never called; mobile data was turned on (toggle ok) but data refuel and excellent speed assertions failed.
- **First failure point:** Planning · **Type:** Planning failure
- **Confound:** Bangla user-simulator error (medium)
- **Evidence:** USER: 'আপনি আমার ডাটা ব্যবহারের পরিমাণ পরীক্ষা করার জন্য কোনো টুল ব্যবহার করতে পারবেন না' then agent: APN check, reboot, transfer.

telecom [service_issue]contract_end_suspension|unseat_sim_card[PERSONA:Hard] · gemma4:31b · C6

- **Task goal:** Your phone has been showing 'No Service' for the past few hours.
- **English:** Agent found line L1002 suspended with contract end date in the past, cited policy that it cannot lift this, and transferred to a human; passes (end state no_service, transfer action).
- **Bangla:** Agent checked L1003 and bills, blamed an overdue bill, sent a payment request, user paid, agent called resume_line despite the expired contract (2025-01-31), then fixed the SIM re-seat; line resumed and service restored, never transferred, so required end state (no_service, DB unchanged) and transfer action failed.
- **First divergence:** At step 7 the BN agent attributed the suspension to an unpaid bill instead of the expired contract date visible in L1002 details, whereas EN reasoned from the contract end date.
- **Final failed / missing action:** transfer_to_human_agents missing; resume_line wrongly executed (policy violation), so db_check and assert_service_status(no_service) failed.
- **First failure point:** Tool selection · **Type:** Wrong tool/action
- **Confound:** General model weakness present in both languages (medium)
- **Evidence:** [ACT #12] resume_line({'customer_id':'C1001','line_id':'L1002'}) with contract_end_date 2025-01-31 in L1002 details; no transfer.

telecom [mms_issue]bad_network_preference|data_mode_off|user_abroad_roaming_disabled_on[PERSONA:None] · gemma4:31b · C2

- **Task goal:** You are unable to send MMS messages using your messaging app for the past few hours. You want to fix it and successfully send an MMS message.
- **English:** User stated up front it is in France; agent saw roaming disabled on L1002, called enable_roaming, then guided user to toggle_data and set 4g_5g_preferred one at a time; MMS worked.
- **Bangla:** BN opening user message omitted that the user is abroad in France (it is in the task's known_info). Agent repeatedly called user-device tools itself (can_send_mms, check_network_status: not found), saw roaming_enabled=false but never acted on it, got user to toggle data and set the mode, then tried APN reset/reboot, Wi-Fi calling, app permissions and transferred to a human.
- **First divergence:** First user turn (France context dropped) and agent never asked about travel/roaming; agent then looked at line details without enabling roaming.
- **Final failed / missing action:** enable_roaming(C1001, L1002) by the agent was never called; assert_can_send_mms failed.
- **First failure point:** Task constraint preservation · **Type:** Dialogue-state failure
- **Confound:** Bangla user-simulator error (low)
- **Evidence:** [USER] হ্যালো, আমি গত কয়েক ঘণ্টা ধরে ... MMS মেসেজ পাঠাতে পারছি না। (no mention of France); get_details_by_id L1002 roaming_enabled:false then no enable_roaming

telecom [mms_issue]bad_network_preference|user_abroad_roaming_disabled_off[PERSONA:None] · gemma4:31b · C1

- **Task goal:** You are unable to send MMS messages using your messaging app for the past few hours. You want to fix it and successfully send an MMS message.
- **English:** User said it was in France in the first message; agent looked up lines, enabled roaming on L1002, user ran toggle_roaming and set_network_mode_preference; MMS worked.
- **Bangla:** User omitted France at first; agent tried device tools itself (not found), asked user to set network mode, APN reset etc. Only after the user mentioned France did agent call enable_roaming for line L1001 (never looked up which line has 555-123-2002; correct line is L1002). The user then claimed to have switched roaming on without any toggle_roaming call appearing. MMS still failed; agent transferred.
- **First divergence:** Agent #15-#16: enable_roaming called with line_id L1001 without any get_details_by_id lookup; earlier the user dropped the France context and the agent never asked.
- **Final failed / missing action:** enable_roaming should be on L1002; user toggle_roaming never actually executed (user text claimed it, no tool call shown).
- **First failure point:** Tool-argument construction · **Type:** Wrong tool argument
- **Confound:** Genuine Bangla-related agent failure (low)
- **Evidence:** enable_roaming({"customer_id": "C1001", "line_id": "L1001"}) vs expected L1002

telecom [mms_issue]break_app_storage_permission|unseat_sim_card|user_abroad_roaming_disabled_on[PERSONA:Easy] · gemma4:31b · C1

- **Task goal:** You are unable to send MMS messages using your messaging app for the past few hours. You want to fix it and successfully send an MMS message.
- **English:** Agent enabled roaming on L1002, then told the user the exact device tools to run (check_network_status, reseat_sim_card, check_app_permissions, grant_app_permission); SIM reseated and storage permission granted; MMS worked.
- **Bangla:** Agent enabled roaming correctly (L1002) but then gave vague natural-language instructions ('make sure Data Roaming is ON') and tried to call device tools itself (check_network_status, toggle_data, check_apn_settings: not found). The simulated user blindly toggled settings (roaming turned OFF, mobile data turned OFF) and never ran reseat_sim_card or grant_app_permission as named tools; agent never found the SIM-missing or storage-permission causes and transferred.
- **First divergence:** Agent #9 and #12: told the user in free text to check/toggle settings instead of naming the exact user tool to run, and called user-side tools itself.
- **Final failed / missing action:** User actions reseat_sim_card and grant_app_permission(messaging, storage) never executed; check_network_status was never obtained from the user.
- **First failure point:** Tool selection · **Type:** Wrong tool/action
- **Confound:** Genuine Bangla-related agent failure (medium)
- **Evidence:** [RESULT] Data Roaming is now OFF. / Error: Tool 'check_network_status' not found.

telecom [mobile_data_issue]airplane_mode_on|bad_network_preference|data_mode_off|data_saver_mode_on[PERSONA:Hard] · gemma4:31b · C1

- **Task goal:** You mobile data is not working properly. It either stops working or is very slow. You want to fix it and absolutely want to get excellent internet speed on your phone. You are not willing to accept any other internet speed (poor, fair or good). You do not have access to wifi.
- **English:** Agent asked the user (hard persona) to run a speed test, check the status bar, turn off airplane mode, enable data, set 4g_5g_preferred, and turn off data saver; speed reached Excellent.
- **Bangla:** Agent tried to run device tools itself (run_speed_test, toggle_airplane_mode, check_network_status, check_network_mode_preference: all 'not found'). Airplane mode was turned off by the user only after a vague text instruction; agent never got data enabled, network mode fixed, or data saver turned off, and after the errors transferred the user to a human. User simulator behaved plausibly (Bangla, anxious persona).
- **First divergence:** Agent #5-#7: offered to run a speed test itself and called run_speed_test as an agent tool.
- **Final failed / missing action:** set_network_mode_preference(4g_5g_preferred), toggle_data and toggle_data_saver_mode were never requested; premature transfer.
- **First failure point:** Tool selection · **Type:** Wrong tool/action
- **Confound:** Genuine Bangla-related agent failure (medium)
- **Evidence:** run_speed_test({}) -> Error: Tool 'run_speed_test' not found. (twice), toggle_airplane_mode({}) -> not found

telecom [mobile_data_issue]bad_network_preference|bad_vpn|data_mode_off|data_saver_mode_on[PERSONA:None] · gemma4:31b · C6

- **Task goal:** You mobile data is not working properly. It either stops working or is very slow. You want to fix it and absolutely want to get excellent internet speed on your phone. You are not willing to accept any other internet speed (poor, fair or good). You do not have access to wifi.
- **English:** After enabling data, setting 4g_5g_preferred and disconnecting VPN gave only 'Good', agent used the Data Saver icon/check_data_restriction_status, user turned data saver off and speed became Excellent.
- **Bangla:** Same three steps completed (data on, 4g_5g_preferred, VPN disconnected) with one hiccup (user entered '4G/5G Preferred', rejected, then succeeded). Speed was 'Good'; agent ignored the Data Saver icon in the status bar, never checked data restriction status, went to data usage and a reboot, then transferred to a human claiming all settings were correct.
- **First divergence:** Agent #15 (speed 'Good'): moved to plan data usage/reboot instead of checking Data Saver as EN did.
- **Final failed / missing action:** toggle_data_saver_mode was never requested; speed stayed at 55 Mbps (Good).
- **First failure point:** Result verification · **Type:** Planning failure
- **Confound:** General model weakness present in both languages (medium)
- **Evidence:** "আপনার ফোনের সমস্ত সেটিংস এখন সঠিক আছে" while status bar showed 🔽 Data Saver

telecom [mobile_data_issue]bad_network_preference|bad_vpn|user_abroad_roaming_disabled_on[PERSONA:None] · gemma4:31b · C2

- **Task goal:** You mobile data is not working properly. It either stops working or is very slow. You want to fix it and absolutely want to get excellent internet speed on your phone. You are not willing to accept any other internet speed (poor, fair or good). You do not have access to wifi.
- **English:** Agent asked whether the user was traveling abroad; user said France; agent enabled roaming on L1002, user disconnected VPN and set 4g_5g_preferred; speed Excellent.
- **Bangla:** BN user never mentioned being abroad in France (known_info) and the agent never asked about travel or checked line L1002 (only get_data_usage on L1001). User fixed network mode and VPN, but the speed test kept failing with 'No Connection'; agent went through data usage, APN and reboot, then transferred to a human.
- **First divergence:** First user turn (France context dropped) combined with agent not asking about location/roaming or calling get_details_by_id for L1002.
- **Final failed / missing action:** enable_roaming(C1001, L1002) never called; mobile data/speed assertions failed.
- **First failure point:** Task constraint preservation · **Type:** Dialogue-state failure
- **Confound:** Bangla user-simulator error (low)
- **Evidence:** [USER] হ্যালো, আমার মোবাইল ডেটা ঠিকমতো কাজ করছে না ... (no mention of France); get_data_usage({"customer_id": "C1001", "line_id": "L1001"})

telecom [mobile_data_issue]bad_network_preference|data_saver_mode_on|data_usage_exceeded[PERSONA:Hard] · gemma4:31b · C6

- **Task goal:** You mobile data is not working properly. It either stops working or is very slow. You want to fix it and absolutely want to get excellent internet speed on your phone. You are not willing to accept any other internet speed (poor, fair or good). You do not have access to wifi.
- **English:** EN: agent asked user to run speed test/status bar, then set_network_mode_preference and toggle_data_saver_mode on device, found 15.1/15 GB, refuel_data 2GB; speed excellent; pass.
- **Bangla:** BN: Hard-persona user said it does not know speed test and asked the agent to do it. Agent called device tools itself (run_speed_test, check_network_status, check_status_bar: not found), never guided the user to set network mode/toggle data saver; refueled 2GB, then transferred to human.
- **First divergence:** Right after user declined to run speed test (BN turn 6) the agent switched from guiding device actions to trying device tools itself and later gave up/transferred.
- **Final failed / missing action:** Missing user-side set_network_mode_preference(4g_5g_preferred) and toggle_data_saver_mode; agent did not persist in guiding user.
- **First failure point:** Planning · **Type:** Planning failure
- **Confound:** General model weakness present in both languages (low)
- **Evidence:** আপনিই দয়া করে আমার হয়ে পরীক্ষা করে দেখুন ... [ACT] run_speed_test({}) -> Tool 'run_speed_test' not found

telecom [service_issue]airplane_mode_on|break_apn_settings|contract_end_suspension|lock_sim_card_pin[PERSONA:None] · gemma4:31b · C2

- **Task goal:** Your phone has been showing 'No Service' for the past few hours.
- **English:** EN: agent checked L1002 (suspended, contract ended) and called transfer_to_human_agents; pass.
- **Bangla:** BN: agent troubleshot device (airplane mode off, SIM PIN locked), never checked line status; asked 'shall I transfer?' and the user answered 'yes ... ###TRANSFER###' in the same message so the agent never called transfer_to_human_agents.
- **First divergence:** User turn 'হ্যাঁ, দয়া করে আমাকে টেকনিক্যাল সাপোর্ট এজেন্টের কাছে ট্রান্সফার করুন। ###TRANSFER###' ended the run before the required tool call.
- **Final failed / missing action:** transfer_to_human_agents (required assistant action) never called; line contract-end suspension never discovered.
- **First failure point:** Premature termination · **Type:** Premature abandonment
- **Confound:** Bangla user-simulator error (medium)
- **Evidence:** হ্যাঁ, দয়া করে আমাকে টেকনিক্যাল সাপোর্ট এজেন্টের কাছে ট্রান্সফার করুন। ###TRANSFER###

telecom [service_issue]airplane_mode_on|break_apn_settings|overdue_bill_suspension|unseat_sim_card[PERSONA:None] · gemma4:31b · C6

- **Task goal:** Your phone has been showing 'No Service' for the past few hours.
- **English:** EN: paid overdue bill, resume_line, then kept troubleshooting (airplane mode, SIM reseat, check_apn, reset_apn, reboot); signal restored; pass.
- **Bangla:** BN: same early steps (airplane off, SIM reseat, bill paid, resume_line) but after the post-resume reboot still showed No Signal the agent transferred to human without checking/resetting APN or rebooting after APN reset.
- **First divergence:** After first reboot failure the BN agent chose transfer instead of continuing APN diagnosis (EN continued); trajectories otherwise near-identical, suggests run-to-run variation.
- **Final failed / missing action:** Missing user-side reset_apn_settings and final reboot_device; transferred prematurely.
- **First failure point:** Premature termination · **Type:** Premature abandonment
- **Confound:** General model weakness present in both languages (low)
- **Evidence:** আমি আমার সাধ্যমতো সব পদক্ষেপ চেষ্টা করেছি ... এখন আমি আপনাকে একজন টেকনিক্যাল সাপোর্ট এজেন্টের কাছে ট্রান্সফার করছি

telecom [service_issue]airplane_mode_on|break_apn_settings|unseat_sim_card[PERSONA:None] · gemma4:31b · C2

- **Task goal:** Your phone has been showing 'No Service' for the past few hours.
- **English:** EN: agent had user check status bar, SIM status, reseat SIM, reset APN, reboot via device tools; signal restored; pass.
- **Bangla:** BN: agent tried device tools itself (check_status_bar, toggle_airplane_mode, check_sim_status, check_apn_settings: not found), user said 'I reseated the SIM' with no reseat tool call, agent misread L1001 and sent payment request for B1002, then transferred after APN reset+reboot still no signal.
- **First divergence:** BN user claimed SIM reseat done without executing reseat_sim_card, and agent never explicitly directed device tool use; reseat_sim_card action unmatched.
- **Final failed / missing action:** reseat_sim_card never executed; unnecessary payment of B1002; premature transfer.
- **First failure point:** Tool selection · **Type:** Wrong tool/action
- **Confound:** Bangla user-simulator error (medium)
- **Evidence:** আমি সিম কার্ডটি পুনরায় লাগিয়ে দেখেছি, কিন্তু তাতেও কোনো কাজ হয়নি।

telecom [service_issue]break_apn_settings|lock_sim_card_pin|overdue_bill_suspension[PERSONA:Easy] · gemma4:31b · C2

- **Task goal:** Your phone has been showing 'No Service' for the past few hours.
- **English:** EN: agent resumed line after bill payment, troubleshooting, then transferred via transfer_to_human_agents; pass.
- **Bangla:** BN: agent found SIM PIN-locked and asked 'shall I transfer you?'; user replied agreeing and appended ###TRANSFER### in the same message, ending run before agent called transfer_to_human_agents.
- **First divergence:** Same-turn assent+###TRANSFER### from BN user simulator before required final tool call.
- **Final failed / missing action:** transfer_to_human_agents (assistant) never called.
- **First failure point:** Premature termination · **Type:** Premature abandonment
- **Confound:** Bangla user-simulator error (medium)
- **Evidence:** হ্যাঁ, দয়া করে আমাকে টেকনিক্যাল সাপোর্ট টিমের কাছে ট্রান্সফার করুন। ... ধন্যবাদ।###TRANSFER###

gpt-oss:20b (26 cases)

airline 8 · gpt-oss:20b · C6

- **Task goal:** You want to book a one-way flight from ORD to PHL on May 26.
- **English:** User sim opened oddly (asking agent for flight number) but cooperated; agent used get_user_details/reservations, found HAT271 and booked correctly.
- **Bangla:** User gave all details up front and said not to ask for DOB; agent never called get_user_details, kept demanding DOB and certificate ID, user switched to English refusing, and conversation ended with no booking.
- **First divergence:** BN agent turn #2 asked for DOB and certificate ID instead of calling get_user_details with the provided user id (EN agent looked up profile).
- **Final failed / missing action:** No get_user_details/search/book_reservation calls at all
- **First failure point:** Planning · **Type:** Planning failure
- **Confound:** General model weakness present in both languages (medium)
- **Evidence:** 'আপনার জন্মতারিখ (DOB) এবং ট্রাভেল সার্টিফিকেটের ID ... ছাড়া বুকিং সম্পন্ন করা সম্ভব নয়' (zero tool calls); user sim also drifted to English

airline 25 · gpt-oss:20b · C1

- **Task goal:** You want to make a reservation for your friend. It should be exactly the same as your current reservation.
- **English:** EN trace is an infrastructure_error placeholder; results.json shows pass (new booking for Ivan Smith made).
- **Bangla:** Agent read the request as modifying the existing 2-passenger reservation down to one passenger, deemed it disallowed and transferred to a human without making the new booking.
- **First divergence:** Agent's first reply framed the request as 'changing your reservation' (passenger list reduction) rather than creating a new booking for Ivan Smith.
- **Final failed / missing action:** Missing book_reservation (DTW-SEA, HAT097/HAT251, Ivan Smith, gift card 128 + credit card 247); agent called transfer_to_human_agents instead.
- **First failure point:** Instruction understanding · **Type:** Understanding failure
- **Confound:** Genuine Bangla-related agent failure (medium)
- **Evidence:** transfer_to_human_agents: 'User wants to reduce number of passengers in a reservation from 2 to 1'; user simulator's last turn also leaked English 'We need to output transfer token.'

airline 49 · gpt-oss:20b · C6

- **Task goal:** You booked the flight and you also purchased insurance for it. You cannot make the flight because you're sick and you want to cancel the flight and get a refund for the flight
- **English:** EN trace file is an infrastructure_error placeholder, but results.json shows reward 1.0 (agent did not cancel 3RK2T9).
- **Bangla:** Agent never looked up the reservation, accepted the user's claim of insurance after one confirmation question, and cancelled 3RK2T9 (insurance actually 'no'), telling the user a full refund was due.
- **First divergence:** BN agent skipped get_reservation_details and trusted the user's insurance claim, going straight to cancel_reservation.
- **Final failed / missing action:** Required: check reservation (get_reservation_details) and not cancel; BN cancelled 3RK2T9 without verifying insurance.
- **First failure point:** Tool selection · **Type:** Verification failure
- **Confound:** General model weakness present in both languages (medium)
- **Evidence:** cancel_reservation({"reservation_id": "3RK2T9"}) right after user 'আমি ফ্লাইটের জন্য বিমা কিনেছি'; result shows insurance: no

banking task_001 · gpt-oss:20b · C7

- **Task goal:** You are playing the role of a customer contacting a customer service representative agent. Your character is a management consultant named Sarah Bosch who earns $100,000 annually. You travel frequently for work, but your company provides a corporate travel card that covers all your work-related travel expenses (flights, hotels, rental cars, etc.). You're looking for a credit card to use for your everyday purchases. Y…
- **English:** Agent searched KB, recommended Gold Rewards Card (2.5%, no fee, needs Rho-Bank+). User then applied on its own via apply_for_credit_card with correct args (Sarah Bosch, 100000, rho_bank_subscription=true); DB matched, reward 1.0.
- **Bangla:** Same flow: KB_search, Bangla recommendation of Gold Rewards Card, user applied for Gold Rewards Card and said ###STOP###. DB check failed: the user-side apply_for_credit_card call did not match the expected args (the call itself is not visible in the trace).
- **First divergence:** Agent turn #3: BN reply told the user the Rho-Bank+ subscription must be activated first (and added unsupported details such as min credit score 720), whereas EN asked the user to check its Rho-Bank+ status; the user's hidden apply call then differed from the expected action.
- **Final failed / missing action:** Required user action apply_for_credit_card(Gold Rewards Card, Sarah Bosch, 100000, rho_bank_subscription=true) did not match in BN (likely a wrong or omitted arg such as rho_bank_subscription or income; not observable).
- **First failure point:** Unclear · **Type:** Unclear
- **Confound:** Unclear (low)
- **Evidence:** [RESULT] Credit card application submitted … / action_match=false for apply_for_credit_card (BN); agent said Rho-Bank+ 'আবশ্যক (প্রথমে সক্রিয় থাকতে হবে)'

retail 0 · gpt-oss:20b · C1

- **Task goal:** You received your order #W2378156 and wish to exchange the mechanical keyboard for the same one but with clicky switches and the smart thermostat for one compatible with Google Home instead of Apple HomeKit. If there is no keyboard that is clicky, RGB backlight, full size, you'd go for no backlight.
- **English:** Agent offered clicky/no-backlight/full-size keyboard 7706410293 and thermostat 7747408585 and exchanged them after user confirmation.
- **Bangla:** Agent offered an 80% clicky no-backlight keyboard (9665000388), dropping the stated full-size requirement; the user accepted and the exchange used the wrong new item.
- **First divergence:** In its first reply the BN agent presented an 80% variant instead of the full-size clicky/no-backlight 7706410293 (same product data seen).
- **Final failed / missing action:** exchange called with new_item_ids [9665000388, ...] instead of [7706410293, 7747408585].
- **First failure point:** Entity or slot grounding · **Type:** Understanding failure
- **Confound:** Genuine Bangla-related agent failure (medium)
- **Evidence:** ACT: exchange_delivered_order_items(new_item_ids=["9665000388","7747408585"]); BN user also invented credit_card_1234567 and an email mismatch (sim noise, secondary)

retail 5 · gpt-oss:20b · C2

- **Task goal:** You want to exchange the water bottle and the desk lamp. You want to exchange the water bottle to a bigger one, and the desk lamp to a less bright one (prefer battery > USB > AC). If the agent asks for confirmation, only exchange the desk lamp. If the agent asks for confirmation again, do not exchange anything, and return the water bottle instead.
- **English:** Agent presented both-item exchange; user said only lamp, then switched to returning the water bottle; return executed (reward 1).
- **Bangla:** BN user revealed id and preferences upfront, said 'yes' to the both-item exchange, so agent executed it; user then broke role (replied in English as if the agent) and later calls failed with 'Non-delivered order' (now exchange requested), ending in transfer.
- **First divergence:** BN user simulator did not follow the 'only desk lamp on first confirmation' script and approved the two-item exchange; agent executed it irreversibly.
- **Final failed / missing action:** no exchange should have occurred; return_delivered_order_items for 8538875209 expected; agent exchanged both items (4579334072, 7453605304) and DB diverged
- **First failure point:** Planning · **Type:** Dialogue-state failure
- **Confound:** Bangla user-simulator error (medium)
- **Evidence:** USER: 'হ্যাঁ।' -> exchange_delivered_order_items 8538875209,8384507844 -> later USER (English): 'I'm sorry for the confusion... Let me update the order' -> Non-delivered order cannot be returned

retail 8 · gpt-oss:20b · C2

- **Task goal:** You want to exchange the water bottle and the desk lamp. You want to exchange the water bottle to a bigger one, and the desk lamp to a brighter one (prefer battery > USB > AC). If the agent asks for confirmation, only exchange the desk lamp.
- **English:** Agent authenticated, presented a summary for both items, user said only the lamp, and the lamp-only exchange was executed (reward 1).
- **Bangla:** BN user dumped everything in the first message including the id as 'name' (and no 'only the lamp' on confirmation); agent skipped a confirmation turn and directly exchanged both bottle and lamp in one call.
- **First divergence:** User simulator gave id as name and revealed all details up front; agent then called exchange_delivered_order_items for both items without asking the user for confirmation, so the 'only the lamp' instruction was never triggered.
- **Final failed / missing action:** exchange should be lamp only (8384507844->9083642334); agent exchanged both items (8538875209->4579334072 also)
- **First failure point:** Planning · **Type:** Verification failure
- **Confound:** Bangla user-simulator error (medium)
- **Evidence:** exchange_delivered_order_items({item_ids:['8538875209','8384507844'], new_item_ids:['4579334072','9083642334'],...}) with no prior confirmation turn

retail 10 · gpt-oss:20b · C2

- **Task goal:** For some reason, you want to return all things ordered. You have two payment methods and two orders, and you want to refund each order to the other order's payment method. If not possible, you are angry and swear for a few times, then ask for human representative. Make sure to return BOTH orders.
- **English:** EN user hallucinated fake order IDs and got frustrated, agent transferred to human; passed by luck (run noise, trajectory not faithful to the task).
- **Bangla:** BN user stated "each order to the other payment method" initially but then mapped each order to its original method; agent did the returns and never transferred.
- **First divergence:** BN user converted the cross-refund goal into original-method refunds when asked for a method per order.
- **Final failed / missing action:** Missing transfer_to_human_agents; returns submitted with original methods.
- **First failure point:** Task constraint preservation · **Type:** Dialogue-state failure
- **Confound:** Bangla user-simulator error (medium)
- **Evidence:** USER: "অর্ডার #W5490111 রিফান্ড হবে credit_card_3124723-এ, #W7387996 paypal_9497703-এ" (originals)

retail 11 · gpt-oss:20b · C2

- **Task goal:** For some reason, you want to return all things ordered. You have two payment methods and two orders, and you want to refund each order to the other order's payment method. If not possible, you are angry and swear for a few times, then agree to return all things with the original payment method.
- **English:** User gave email, agent looked up both orders, user insisted on swapped refunds then accepted originals, agent issued both returns (credit card for #W5490111, PayPal for #W7387996); DB match.
- **Bangla:** Bangla user simulator refused to give the email (which its known_info contains) and instead invented order numbers 876543/987654; agent could not authenticate and transferred to human, no returns made.
- **First divergence:** BN user turn 2 refused to share email/personal info and fabricated order IDs, whereas EN user supplied mia.garcia2723@example.com immediately.
- **Final failed / missing action:** All reads (find_user_id_by_email etc.) and both return_delivered_order_items calls missing; agent transferred instead.
- **First failure point:** Task constraint preservation · **Type:** Dialogue-state failure
- **Confound:** Bangla user-simulator error (medium)
- **Evidence:** USER: "আমি আমার ইমেইল বা ব্যক্তিগত তথ্য শেয়ার করতে চাই না ... অর্ডার নম্বরগুলো হলো 876543 এবং 987654" -> transfer_to_human_agents

retail 12 · gpt-oss:20b · C2

- **Task goal:** You just got into gaming and want to cancel or return everything not associated with it. (Everything except a keyboard and a mouse, but do not reveal it to the agent). PayPal is prefered for refund, but otherwise you are angry and ask for human agent for help.
- **English:** User (leaking its reasoning) gave fake order IDs and asked for a human; agent transferred to human without any write; DB match.
- **Bangla:** User sim switched to English, accepted agent-invented order dates and changed goal to return EVERYTHING in both orders (dropping the keep-keyboard-and-mouse constraint); agent executed two return_delivered_order_items writes and then transferred, so DB mismatched (expected no writes, only transfer).
- **First divergence:** After the agent listed the orders with fabricated approximate dates, BN user asked to return all items from both orders; EN agent just transferred.
- **Final failed / missing action:** Unexpected writes: return_delivered_order_items x2 (expected only transfer_to_human_agents, no DB change).
- **First failure point:** Planning · **Type:** Wrong tool/action
- **Confound:** Bangla user-simulator error (medium)
- **Evidence:** USER: "I'd like to return everything from both orders" ; ACT #10/#11 return_delivered_order_items, then transfer_to_human_agents

retail 14 · gpt-oss:20b · C2

- **Task goal:** You just quit gaming and want to cancel or return everything associated with it. (It's just a keyboard and a mouse, but do not reveal it to the agent). Original payment is preferred.
- **English:** User asked to return gaming-related items; agent looked up both orders and returned keyboard and gaming mouse to original methods.
- **Bangla:** BN user simulator turned the goal into cancelling a recent order and then role-reversed, repeating agent-like questions; agent never fetched user details/orders and the episode ended OUT-OF-SCOPE.
- **First divergence:** First BN user turn stated a different goal (cancel an order) with no mention of gaming/returns.
- **Final failed / missing action:** No get_user_details/get_order_details or return_delivered_order_items calls.
- **First failure point:** Instruction understanding · **Type:** Understanding failure
- **Confound:** Bangla user-simulator error (high)
- **Evidence:** USER: "আমি সম্প্রতি একটি অর্ডার দিয়েছি, কিন্তু এখন আমি সেটি cancel করতে চাই" ... later "###OUT-OF-SCOPE###"; agent claimed it could not see orders

retail 22 · gpt-oss:20b · C6

- **Task goal:** You want to change your user address and all possible order addresses to be 101 Highway, New York, 10001 (including the orders). Once the agent has confirmed the changes for the default address AND the order addresses, then you regret it and want to change the user address back to the original address.
- **English:** Agent updated default address, asked confirmation, fetched all 3 orders, updated the one pending order, then restored original default address (667 Highland Drive, Suite 865, Denver) when user reverted.
- **Bangla:** Agent updated default address and immediately claimed all order addresses would be updated without ever fetching orders or calling modify_pending_order_address; user then asked to revert with only "Denver, 80280" and agent wrote address1="Denver" instead of the real original address.
- **First divergence:** BN agent step #4-#5: right after modify_user_address it said all order addresses would be updated, skipping get_order_details and modify_pending_order_address (EN fetched orders and updated #W9911714).
- **Final failed / missing action:** Missing get_order_details x3 and modify_pending_order_address(#W9911714); final modify_user_address had wrong address1 "Denver" / no Suite 865 / not 667 Highland Drive.
- **First failure point:** Planning · **Type:** Verification failure
- **Confound:** General model weakness present in both languages (medium)
- **Evidence:** ACT #7 modify_user_address({"address1": "Denver", "address2": "", "city": "Denver", ... "zip": "80280"}); no order calls

retail 49 · gpt-oss:20b · C2

- **Task goal:** You mistakenly ordered a Wireless Earbud with an IPX7 water resistance level, but you don't require this feature. You wish to exchange it for one with the same water resistance level as the other Wireless Earbuds that you've purchased. In fact, you want to exchange it to the cheapest earbud item from the rest of that order.
- **English:** (EN trace file is an empty infrastructure_error stub, but results.json shows reward 1.0 with expected actions matched, ending with exchange of earbud 2757705742 to 1646531091.)
- **Bangla:** User sim misstated the goal as exchanging for another IPX7 earbud, gave Latin name variant and hallucinated order contents; agent found #W3470184 and exchanged 2757705742 for 9580569596 (cheapest IPX7 variant, $257.38) instead of cheapest other earbud in the order (1646531091); also fumbled price difference.
- **First divergence:** BN user first turn changed the constraint to "same IPX7 level, cheapest earbud in the order" (original: same water-resistance level as the other earbuds, cheapest item from rest of order); agent then picked 9580569596.
- **Final failed / missing action:** exchange_delivered_order_items new_item_ids should be ["1646531091"] (and calculate(258.97-232.49)); agent used ["9580569596"].
- **First failure point:** Task constraint preservation · **Type:** Dialogue-state failure
- **Confound:** Bangla user-simulator error (medium)
- **Evidence:** ACT #18 exchange_delivered_order_items({"item_ids":["2757705742"],"new_item_ids": ["9580569596"],"order_id":"#W3470184","payment_method_id":"gift_card_7245904"})

retail 53 · gpt-oss:20b · C2

- **Task goal:** The bicycle you received was damaged during delivery, and you want to get a refund. You're quite frustrated because the bike was very expensive and you'd like to receive the refund as soon as possible. You want the refund to be credited to your original credit card.
- **English:** (EN trace file is an empty infrastructure_error stub, but results.json shows reward 1.0 with all expected actions matched: found sofia_li_9219 and returned bicycle in #W3916020 to credit card.)
- **Bangla:** Bangla user sim introduced itself as "সোহাই লি" (mis-rendering of Sofia Li) so name+zip lookups ("সোহাই"/"Sohai") failed; sim invented a phone number, then emitted OUT-OF-SCOPE and agent transferred; no return.
- **First divergence:** BN user first message corrupted the name Sofia into Bengali transliteration "সোহাই", so authentication failed.
- **Final failed / missing action:** find_user_id_by_name_zip with Sofia/Li/78260 and return_delivered_order_items(#W3916020, 7758198585, credit_card_8105988) never made.
- **First failure point:** Entity or slot grounding · **Type:** Dialogue-state failure
- **Confound:** Bangla user-simulator error (high)
- **Evidence:** USER: "আমি সোহাই লি, সান অ্যান্টোনিও, ৭৮২৬০ থেকে" ; find_user_id_by_name_zip("Sohai","Li","78260") -> User not found

telecom [mobile_data_issue]airplane_mode_on|data_saver_mode_on|user_abroad_roaming_disabled_on[PERSONA:None] · gpt-oss:20b · C2

- **Task goal:** You mobile data is not working properly. It either stops working or is very slow. You want to fix it and absolutely want to get excellent internet speed on your phone. You are not willing to accept any other internet speed (poor, fair or good). You do not have access to wifi.
- **English:** Agent found L1002, enabled roaming, user toggled airplane mode/data saver off, speed test 275 Mbps excellent (reward 1.0).
- **Bangla:** The Bangla user simulator switched to English and role-reversed (acting like the agent, asking the agent to confirm toggles), never performed airplane mode/data saver actions properly, loop of repeated requests until it emitted ###OUT-OF-SCOPE###; roaming was enabled but airplane mode stayed ON.
- **First divergence:** From the first replies the user turned English and began giving instructions to the agent ("Could you please toggle Airplane Mode off?") instead of acting as the customer.
- **Final failed / missing action:** user toggle_airplane_mode and toggle_data_saver_mode never executed (action_match false); airplane mode stayed ON; env assertions failed
- **First failure point:** Instruction understanding · **Type:** Dialogue-state failure
- **Confound:** Bangla user-simulator error (high)
- **Evidence:** [USER] "It looks like your phone is currently in Airplane Mode... Could you please toggle Airplane Mode off?" then loops "Airplane Mode still showing as ON"

telecom [mobile_data_issue]bad_network_preference|bad_vpn|data_usage_exceeded|user_abroad_roaming_disabled_off[PERSONA:Easy] · gpt-oss:20b · C6

- **Task goal:** You mobile data is not working properly. It either stops working or is very slow. You want to fix it and absolutely want to get excellent internet speed on your phone. You are not willing to accept any other internet speed (poor, fair or good). You do not have access to wifi.
- **English:** Agent identified L1002 via phone 555-123-2002, enabled roaming, refueled 2 GB, user toggled roaming, switched network mode and disconnected VPN; Excellent speed.
- **Bangla:** Agent assumed phone 555-123-2002 was line L1001 without verifying, checked L1001 usage (under limit), toggled roaming on/off on L1001, never inspected L1002 so never refueled/enabled roaming there; user fixed network mode but VPN and roaming/refuel remained, ended in transfer.
- **First divergence:** At turn 6 the BN agent declared '555-123-2002 is L1001' after one get_data_usage on L1001, while EN fetched L1002 details by id and found the 15.1 GB overage.
- **Final failed / missing action:** refuel_data and enable_roaming on L1002 missing; disconnect_vpn not performed by user; enable then disable_roaming on wrong line L1001 (user-side toggle_roaming match true).
- **First failure point:** Entity or slot grounding · **Type:** Understanding failure
- **Confound:** General model weakness present in both languages (medium)
- **Evidence:** '৫৫৫-১২৩-২০০২ নম্বরের লাইনটি L1001। আপনার ডেটা ব্যবহার ৩.২ GB' ; ACT enable_roaming L1001 then disable_roaming L1001. Note BN user sim also altered name to 'জোন স্যামসন', used Bengali digits and role-reversed.

telecom [mobile_data_issue]bad_network_preference|user_abroad_roaming_enabled_off[PERSONA:Hard] · gpt-oss:20b · C2

- **Task goal:** You mobile data is not working properly. It either stops working or is very slow. You want to fix it and absolutely want to get excellent internet speed on your phone. You are not willing to accept any other internet speed (poor, fair or good). You do not have access to wifi.
- **English:** User simulator toggled roaming and set network mode 4g_5g_preferred via device tools, speed test hit 275 Mbps, reward 1.0.
- **Bangla:** After the first Bangla turn the user simulator switched to English, role-reversed (wrote agent-style instructions) and only claimed to change settings, so network mode stayed 2G_only and speed never reached excellent; agent eventually transferred to a human.
- **First divergence:** At the first agent instruction the BN user sim answered in prose (claimed 'I'm changing network mode') instead of calling set_network_mode_preference, and later turns were in English role-reversal; the first user turn also added an unprompted 2 GB refuel willingness.
- **Final failed / missing action:** User-side set_network_mode_preference(4g_5g_preferred) never executed (action_match false); assert_internet_speed excellent unmet (toggle_roaming did happen).
- **First failure point:** Unclear · **Type:** Dialogue-state failure
- **Confound:** Bangla user-simulator error (medium)
- **Evidence:** [USER] 'I'm sorry, but I don't have a tool that can reset the network settings directly.' ; final network status still 'Cellular Network Type: 2G'

telecom [mobile_data_issue]data_usage_exceeded|user_abroad_roaming_enabled_off[PERSONA:Easy] · gpt-oss:20b · C6

- **Task goal:** You mobile data is not working properly. It either stops working or is very slow. You want to fix it and absolutely want to get excellent internet speed on your phone. You are not willing to accept any other internet speed (poor, fair or good). You do not have access to wifi.
- **English:** Agent refueled 2 GB on L1002 (user asked for 2 GB), user turned roaming on, speed test excellent, env assertions met (reward 1.0).
- **Bangla:** Agent first called enable_roaming on the wrong line L1001, then told the user max refuel is 2 GB but suggested 0.9 GB as an example; user picked 1 GB, refuel_data(gb_amount=1) failed assert_data_refueling_amount=2.0 although speed became excellent.
- **First divergence:** BN agent opened with enable_roaming on L1001 and a 0.9/1 GB refuel suggestion, whereas EN agent asked how much and the user answered 2 GB.
- **Final failed / missing action:** refuel_data should be 2.0 GB on L1002; BN did 1 GB (assert_data_refueling_amount false). Roaming also enabled on wrong line L1001.
- **First failure point:** Tool-argument construction · **Type:** Wrong tool argument
- **Confound:** General model weakness present in both languages (medium)
- **Evidence:** enable_roaming({customer_id:C1001,line_id:L1001}); refuel_data({line_id:L1002,gb_amount:1}); user: 'আমি ১ GB রিফুয়েল করতে চাই'

telecom [mobile_data_issue]bad_network_preference|bad_vpn|data_mode_off|data_saver_mode_on|data_usage_exceeded|user_abroad_roaming_enabled_off[PERSON · gpt-oss:20b · C2

- **Task goal:** You mobile data is not working properly. It either stops working or is very slow. You want to fix it and absolutely want to get excellent internet speed on your phone. You are not willing to accept any other internet speed (poor, fair or good). You do not have access to wifi.
- **English:** EN trace file is an infrastructure_error stub (empty); results.json for the EN task shows reward 1.0 with all 6 actions and 3 assertions met.
- **Bangla:** Agent refueled 2 GB correctly but the Bangla user simulator role-reversed, never used device tools, asked the agent to run the speed test and looped ~25 turns until it emitted ###OUT-OF-SCOPE###; device state (mode 2G, VPN, data saver, roaming) never fixed.
- **First divergence:** From the first agent guidance turn the BN user sim answered with prose/agent-like tables and later English, instead of performing set_network_mode/disconnect_vpn/toggle_data_saver/toggle_roaming; agent also fabricated a roaming tool call in text.
- **Final failed / missing action:** User actions disconnect_vpn, toggle_data_saver_mode, toggle_roaming not performed; assert_mobile_data_status and assert_internet_speed unmet (refuel assertion met).
- **First failure point:** Unclear · **Type:** Dialogue-state failure
- **Confound:** Bangla user-simulator error (high)
- **Evidence:** [USER] 'I'm sorry, but I can't run Speedtest-Net myself. Could you please open the app on your phone...' ; final [USER] ###OUT-OF-SCOPE###

telecom [mobile_data_issue]airplane_mode_on|bad_network_preference|data_mode_off|data_saver_mode_on[PERSONA:Hard] · gpt-oss:20b · C6

- **Task goal:** You mobile data is not working properly. It either stops working or is very slow. You want to fix it and absolutely want to get excellent internet speed on your phone. You are not willing to accept any other internet speed (poor, fair or good). You do not have access to wifi.
- **English:** Agent asked for DOB, name lookup returned nothing, user gave phone 555-123-2002, lookup succeeded, and agent walked the user through toggle_data, data saver, network mode, airplane mode; speed Excellent.
- **Bangla:** Agent asked for DOB, then called get_customer_by_phone with the placeholder '?', then asked for a phone number with an example format '+1-555-123-4567'. The user answered '+1-555-123-2002', the lookup failed, and the agent transferred to a human after the user could not give DOB/customer ID, never retrying the number without +1 or a name lookup. No device troubleshooting happened. (User simulator replied in English, as in all BN user turns.)
- **First divergence:** Agent #3-#5: lookup with phone '?' and a +1 format example that led to a non-matching number; then early transfer at #8.
- **Final failed / missing action:** Never authenticated the customer; never guided airplane mode off, network mode, data on, data saver off.
- **First failure point:** Tool-argument construction · **Type:** Wrong tool argument
- **Confound:** General model weakness present in both languages (medium)
- **Evidence:** get_customer_by_phone({"phone_number": "+1-555-123-2002"}) -> Error: Customer with phone number +1-555-123-2002 not found

telecom [mobile_data_issue]bad_network_preference|bad_vpn|data_mode_off|data_saver_mode_on[PERSONA:None] · gpt-oss:20b · C6

- **Task goal:** You mobile data is not working properly. It either stops working or is very slow. You want to fix it and absolutely want to get excellent internet speed on your phone. You are not willing to accept any other internet speed (poor, fair or good). You do not have access to wifi.
- **English:** Agent diagnosed 2G, mobile data off, VPN and data saver, and guided the user through set_network_mode_preference, toggle_data, disconnect_vpn, toggle_data_saver_mode; speed Excellent.
- **Bangla:** User (Bangla) gave number up front. Agent got data on, 4g_5g_preferred, an unneeded toggle_roaming, data saver off, but VPN stayed on. After the user said 'VPN is still on', the agent wrote as if the VPN had been turned off, never asked for disconnect_vpn, and spent turns on APN, firmware, power-saving and carrier support. The user mildly hedged about keeping the VPN but said turning it off was fine. User said ###STOP### at speed 27.5 Mbps (Good).
- **First divergence:** Agent #12: misread the user report ('এখন VPN বন্ধ করে 27.5 Mbps') while VPN was still connected, instead of requesting disconnect_vpn.
- **Final failed / missing action:** disconnect_vpn never executed (ground truth action_match false); speed not Excellent.
- **First failure point:** Result verification · **Type:** Verification failure
- **Confound:** General model weakness present in both languages (medium)
- **Evidence:** ডেটা সেভার বন্ধ করেছি, এবং এখন VPN এখনও চালু আছে ... VPN: চালু ও সংযুক্ত

telecom [mobile_data_issue]bad_network_preference|bad_vpn|user_abroad_roaming_disabled_on[PERSONA:None] · gpt-oss:20b · C2

- **Task goal:** You mobile data is not working properly. It either stops working or is very slow. You want to fix it and absolutely want to get excellent internet speed on your phone. You are not willing to accept any other internet speed (poor, fair or good). You do not have access to wifi.
- **English:** EN: agent enabled roaming on L1002, user ran status bar/set_network_mode_preference/disconnect_vpn via tools with real results; speed test 275 Mbps excellent; pass.
- **Bangla:** BN: agent gave a long generic tool-name checklist; BN user replied in English and then narrated device actions (APN reset, reboot, speed test 'Excellent') with no device tool calls. VPN never disconnected; agent accepted this and closed. Env assertion (excellent speed) failed.
- **First divergence:** After agent turn 13 the BN user stopped grounding replies in tool results and fabricated APN/speed-test outcomes in English; agent never asked for status bar so never learned VPN was on.
- **Final failed / missing action:** User-side disconnect_vpn never executed; speed assertion not met; agent also enabled roaming on L1001 needlessly.
- **First failure point:** Result verification · **Type:** Dialogue-state failure
- **Confound:** Bangla user-simulator error (medium)
- **Evidence:** I followed the steps you outlined: ... Ran `run_speed_test()` ... the result is “Excellent” internet speed.

telecom [mobile_data_issue]bad_vpn|data_mode_off|data_usage_exceeded|user_abroad_roaming_disabled_off[PERSONA:None] · gpt-oss:20b · C2

- **Task goal:** You mobile data is not working properly. It either stops working or is very slow. You want to fix it and absolutely want to get excellent internet speed on your phone. You are not willing to accept any other internet speed (poor, fair or good). You do not have access to wifi.
- **English:** EN: also used L1001 at first, but user ran device tools with real outputs; agent later found L1002, refueled 2GB, user disconnected VPN; excellent speed; pass.
- **Bangla:** BN: agent never looked at L1002 (the 2002 line); enabled roaming/queried data on L1001 only. BN user often replied in English, narrated steps ('checked and reset APN, verified SIM...') without tool calls, invented factory reset/SIM replacement; ~60 turns, ended with transfer.
- **First divergence:** Agent mapped 555-123-2002 to L1001 (also in EN) and BN user simulator then supplied ungrounded, fabricated device results so agent never recovered.
- **Final failed / missing action:** refuel_data and enable_roaming on L1002, toggle_roaming/toggle_data, disconnect_vpn never executed; wrong line used; unnecessary send_payment_request for B1002.
- **First failure point:** Entity or slot grounding · **Type:** Dialogue-state failure
- **Confound:** Bangla user-simulator error (medium)
- **Evidence:** I’ve followed every step you listed—checked and reset the APN, verified the SIM is active ... Still, the speed test comes back “No Connection.”

telecom [service_issue]airplane_mode_on|break_apn_settings|unseat_sim_card[PERSONA:None] · gpt-oss:20b · C2

- **Task goal:** Your phone has been showing 'No Service' for the past few hours.
- **English:** EN: user ran check_network_status, reseat_sim_card, airplane off, check_apn and reset_apn/reboot via device tools; signal restored; pass.
- **Bangla:** BN: agent asked which line; BN user answered with internal ID 'L1001' (phone is 2002/L1002). Agent gave prose SIM-reseat steps; user narrated reseating without calling reseat_sim_card, SIM stayed missing, agent never reached APN and transferred.
- **First divergence:** User simulator surfaced internal line ID and then described physical actions instead of executing device tools; reseat/reset_apn/reboot never run.
- **Final failed / missing action:** reseat_sim_card, reset_apn_settings, reboot_device never executed by user; agent transferred early.
- **First failure point:** Entity or slot grounding · **Type:** Dialogue-state failure
- **Confound:** Bangla user-simulator error (medium)
- **Evidence:** আমি L1001 লাইনটি ব্যবহার করছি।

telecom [service_issue]break_apn_settings|lock_sim_card_pin[PERSONA:None] · gpt-oss:20b · C2

- **Task goal:** Your phone has been showing 'No Service' for the past few hours.
- **English:** EN: agent checked status bar and SIM status (locked PIN) then called transfer_to_human_agents; pass.
- **Bangla:** BN: agent saw locked_pin but gave DIY PIN/PUK instructions instead of transferring; BN user reversed roles ('please write your SIM PIN, I will guide you'); agent apologised, ended chat, user emitted ###STOP### with no transfer call.
- **First divergence:** At PIN-locked result agent did not transfer; user role-reversal then STOP ended run without required transfer.
- **Final failed / missing action:** transfer_to_human_agents never called.
- **First failure point:** Premature termination · **Type:** Premature abandonment
- **Confound:** Bangla user-simulator error (low)
- **Evidence:** অবশ্যই! দয়া করে আপনার SIM-এর PIN লিখে দিন, আমি আপনাকে কীভাবে তা প্রবেশ করতে হবে তা নির্দেশনা দেব।

telecom [service_issue]overdue_bill_suspension|unseat_sim_card[PERSONA:Easy] · gpt-oss:20b · C2

- **Task goal:** Your phone has been showing 'No Service' for the past few hours.
- **English:** EN: pay bill, resume_line, reboot, then user reseated SIM via tool; signal 5G excellent; pass.
- **Bangla:** BN: bill paid and line resumed fine; SIM missing. Agent tried reseat_sim_card itself (not found) and gave prose steps; BN user said 'I am reseating now' but only re-ran check_sim_status (still no SIM) repeatedly; agent transferred.
- **First divergence:** BN user narrated reseating without executing reseat_sim_card, so SIM never seated; service assertion false.
- **Final failed / missing action:** User-side reseat_sim_card not actually executed; agent transferred instead of explicitly directing the device tool.
- **First failure point:** Result verification · **Type:** Dialogue-state failure
- **Confound:** Bangla user-simulator error (medium)
- **Evidence:** ঠিক আছে, আমি এখন SIM-কার্ডটি বের করে আবার সঠিকভাবে বসিয়ে দিচ্ছি ... [RESULT] No SIM card detected in the phone.

Appendix B. Comparison groups

Automatic signal	gemma PF (BN)	gemma FF (BN)	gemma FF (EN)	gpt-oss PF (BN)	gpt-oss FF (BN)	gpt-oss FF (EN)
Simulator ended before the agent acted	17	29	23	0	3	3
Required write never called	6	31	36	11	32	28
Write called with wrong arguments	4	15	18	4	17	22
User-side device action never done	7	17	14	7	6	6
Extra or wrong change in database	5	3	3	3	5	4
Required info not told to user	1	0	0	0	0	0
User-side device state not reached	0	0	1	1	1	1
Out of steps	0	0	0	0	0	0
Other	0	0	0	0	0	0

Model	Group	Airline	Retail	Telecom	Banking	Total
gemma4:31b	EN-pass / BN-pass	21	19	25	3	68
	EN-pass / BN-fail	12	11	16	1	40
	EN-fail / BN-pass	2	2	11	0	15
	EN-fail / BN-fail	15	9	35	36	95
gpt-oss:20b	EN-pass / BN-pass	16	8	6	2	32
	EN-pass / BN-fail	3	10	12	1	26
	EN-fail / BN-pass	6	3	17	1	27
	EN-fail / BN-fail	23	20	11	10	64